

CIP-B106: Mobile and IoT Forensics

iPhone iOS 13.4.1 forensic image

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Date: [4 October, 2026]

Platform Used

Kali Linux (VMware Workstation)

C11-26-DFIT17314_CIP-B106_22092026

CIP-B106_-26-DFIT17314_FAVOUR_MADU_22092026

INSTRUCTOR: AMINU IDRIS, AMCPN | FOUNDER, ICDFA

Academic Integrity and Authorization Statement

I **Favour Madu** confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.

Executive Summary

This report documents a forensic examination of an iPhone image running iOS 13.4.1 using Kali Linux in a VMware Workstation environment. The work began with acquisition of the published extraction, verification of file integrity using cryptographic hashes, and examination of the extracted iOS file-system structure. Attention was given to application bundles, user-data containers, property-list files and SQLite databases that commonly contain evidence relevant to mobile-device investigations.

The examination identified device and subscriber information, including the registered telephone number, ICCID, IMSI, IMEI, and mobile network codes. It also mapped key artefact locations for Messages, contacts, call history, voicemail, calendars, Safari, photos, KnowledgeC, Health, and location services. Review of Safari history established that the device was used to search for information about the 2020 MLB season and the resumption of the NHL between approximately 00:58 and 01:02 GMT on 28 March 2020. No voicemail message was identified in the examined voicemail data.

Overall, the exercise demonstrates how native iOS artefacts can be located, interpreted, and correlated to reconstruct device configuration and selected user activity. The results show the evidential value of combining file-system navigation, property-list parsing, SQLite analysis, timestamp conversion, and integrity verification. Several later sections record investigative workflows or database locations without presenting a complete evidential conclusion; accordingly, any operational use of those areas should include validation of the underlying records and preservation of a clear chain of custody.

Key Recommendations

- Preserve the original forensic image as read-only and conduct all analysis on verified working copies, recording cryptographic hashes before and after transfer or extraction.
- Maintain a complete chain of custody and examination log covering evidence handling, tool versions, commands, time-zone assumptions, and any processing errors.
- Validate significant findings with at least one independent method or forensic tool, particularly parsed property-list values, SQLite records, deleted data, and converted Apple timestamps.
- Complete the unresolved artefact examinations—especially calendar, call-history, contacts, photos, KnowledgeC, Health, location, account, device model, serial-number, and jailbreak indicators—and state a clear finding for each area.
- Correlate related artefacts across databases and file-system sources before drawing conclusions about user activity; distinguish observed facts from interpretation and document evidential confidence.
- Protect sensitive identifiers and personal data in shared or assessed copies by redacting values where disclosure is unnecessary, while retaining an access-controlled evidential version.
- Export relevant query results and screenshots with clear labels, source paths, and timestamps to ensure each conclusion can be independently reproduced and verified.

INTRODUCTION TO iOS 13 IMAGE

- iOS Image Description
- File System Structure
- Understanding the Application Bundles

iOS 13 IMAGE DESCRIPTION -Created by Joshua Hickman- Associate with Team Win Recovery Project (TWRP)

Created via Magnet Acquire

Running iOS 13.4.1

All files are included in the zip file iOS 13.4.1 Extraction.zip. Once extracted multiple files are present.

13.4.1.tar (Extraction)

iTunes Backup

Sysdiagnose Logs

link to download the lab

wget -q https://digitalcorpora.s3.amazonaws.com/corpora/mobile/ios_13_4_1/ios_13_4_1.zip

CREATED A DIRECTORY on linux -a virtual environment- mkdir iOS_13

Download the zip file

```
(kali@kali)~$ mkdir iOS_13
(kali@kali)~$ cd iOS_13
(kali@kali)~/iOS_13$ wget -q https://digitalcorpora.s3.amazonaws.com/corpora/mobile/ios_13_4_1/ios_13_4_1.zip
Prepended http:// to '-q'
--2026-02-21 12:51:49-- http://xn--q-5gn/
Resolving xn--q-5gn (xn--q-5gn)... failed: Name or service not known.
wget: unable to resolve host address 'xn--q-5gn'
--2026-02-21 12:52:05-- https://digitalcorpora.s3.amazonaws.com/corpora/mobile/ios_13_4_1/ios_13_4_1.zip
Resolving digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)... 52.92.185.25, 3.5.81.8, 52.92.224.41, ...
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)|52.92.185.25|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9629286492 (9.0G) [application/zip]
Saving to: 'ios_13_4_1.zip'

ios_13_4_1.zip          99%[=====] 8.96G 2.71MB/s  in 62m 50s

2026-02-21 13:54:56 (2.43 MB/s) - Read error at byte 9624024646/9629286492 (Error decoding the received TLS packet.). Retrying.
--2026-02-21 13:54:57-- (try: 2) https://digitalcorpora.s3.amazonaws.com/corpora/mobile/ios_13_4_1/ios_13_4_1.zip
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)|52.92.185.25|:443... connected.
HTTP request sent, awaiting response... 206 Partial Content
Length: 9629286492 (9.0G), 5261846 (5.0M) remaining [application/zip]
Saving to: 'ios_13_4_1.zip'

ios_13_4_1.zip          100%[+++++] 8.97G 1.45MB/s  in 3.5s

2026-02-21 13:55:01 (1.45 MB/s) - 'ios_13_4_1.zip' saved [9629286492/9629286492]

FINISHED --2026-02-21 13:55:01--
Total wall clock time: 1h 3m 11s
Downloaded: 1 files, 5.0M in 1h 2m 53s (1.36 KB/s)
```

Confirm the full download of the file.-using `ls ios_13_4_1.zip -l`

```
(kali@kali)~/iOS_13$ ls ios_13_4_1.zip -l
-rw-rw-r-- 1 kali kali 9629286492 Nov 21  2020 ios_13_4_1.zip

(kali@kali)~/iOS_13$ hashdeep ios_13_4_1.zip
%%%% HASHDEEP-1.0
%%%% size,md5,sha256,filename
## Invoked from: /home/kali/iOS_13
## $ hashdeep ios_13_4_1.zip
##
9629286492,4e094801ba1e3b938b2c0cc18d0e0d1d,761854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c154551,/home/kali/iOS_13/ios_13_4_1.zip
```

VERIFY HASHES

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Hashdeep the unzip file to verify the integrity of the file - **hashdeep ios_13_4_1.zip**

MD5: 4e094801ba1e3b938b2c0cc18d0e0d1d

SHA-256: 761854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c15551

9629286492,4e094801ba1e3b938b2c0cc18d0e0d1d,761854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c154551,

Unzip to get iOS 13 extraction

Unzip the file- **unzip -q ios_13_4_1.zip**

List the content of the file- **ls -l**

```
(kali@kali)-[~/ios_13]
$ unzip -q ios_13_4_1.zip

(kali@kali)-[~/ios_13]
$ ls -l
total 9403612
drwxr-xr-x 6 kali kali      4096 Apr 19  2020 'iOS 13.4.1 Extraction'
-rw-rw-r-- 1 kali kali 9629286492 Nov 21  2020 ios_13_4_1.zip
```

Unzip tar - **tar -xvf 13-4-1.tar**

```
(kali@kali)-[~/ios_13/iOS 13.4.1 Extraction/Extraction]
$ tar -xvf 13-4-1.tar
Volumes/JOSH/NoTar-13-4-1/
Volumes/JOSH/NoTar-13-4-1/.ba/
Volumes/JOSH/NoTar-13-4-1/.file
Volumes/JOSH/NoTar-13-4-1/.mb/
Volumes/JOSH/NoTar-13-4-1/.Trashes
Volumes/JOSH/NoTar-13-4-1/Applications/
Volumes/JOSH/NoTar-13-4-1/bin/
Volumes/JOSH/NoTar-13-4-1/cores/
Volumes/JOSH/NoTar-13-4-1/dev/
Volumes/JOSH/NoTar-13-4-1/Developer/
Volumes/JOSH/NoTar-13-4-1/etc
Volumes/JOSH/NoTar-13-4-1/Library/
```

FILE SYSTEM STRUCTURE

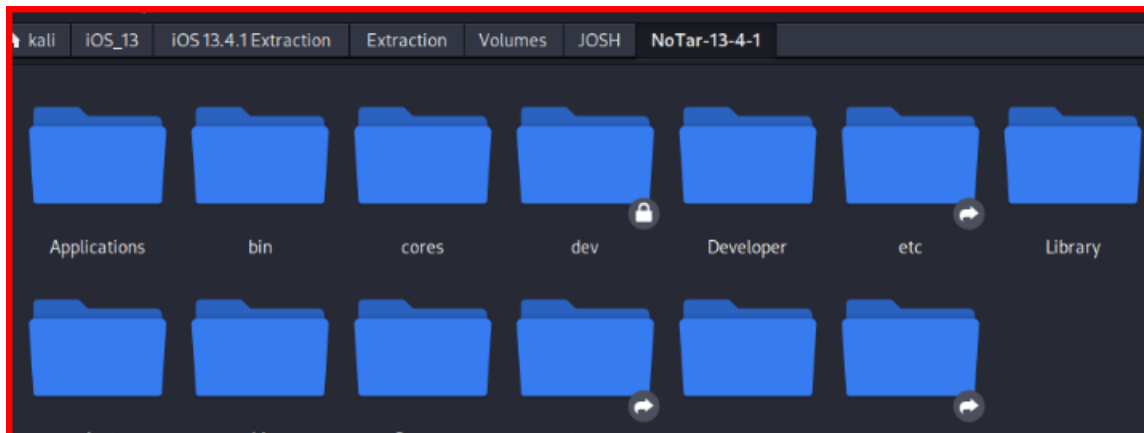
cd ios_13

cd iOS\ 13.4.1\ Extraction\Extraction\13-4-1\Volumes\JOSH\NoTar-13-4-1

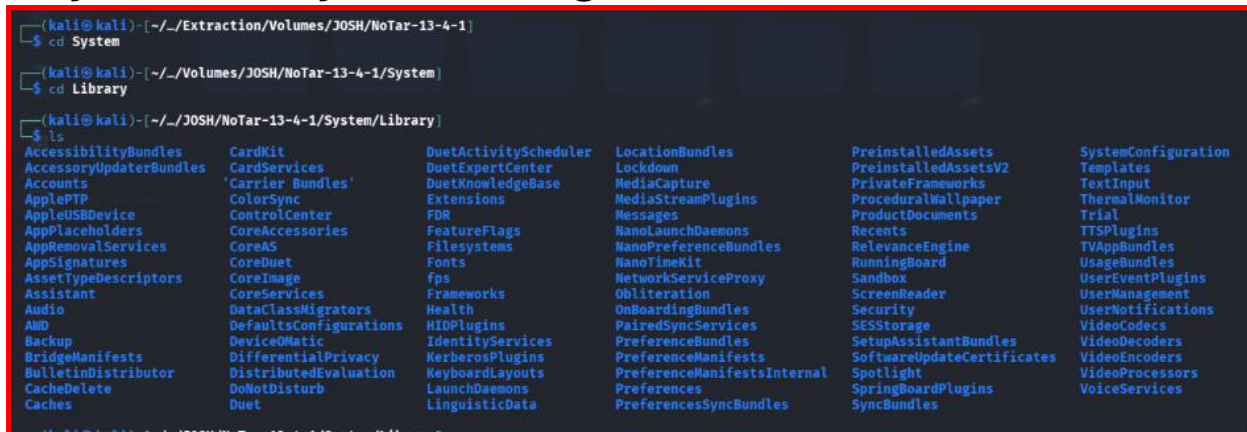
ls -l

```
(kali@kali)-[~/Extraction/Volumes/JOSH/NoTar-13-4-1]
$ ls -l
total 40
drwxrwxr-x 2 kali kali 4096 Feb 21 17:19 Applications
drwxr-xr-x 2 kali kali 4096 Feb 21 17:19 bin
drwxrwxr-x 2 kali kali 4096 Apr 2  2020 cores
dr-xr-xr-x 2 kali kali 4096 Apr 16  2020 dev
drwxrwxr-x 2 kali kali 4096 Apr 2  2020 Developer
lrwxrwxrwx 1 kali kali 11 Apr 2  2020 etc -> private/etc
drwxrwxr-x 2 kali kali 4096 Feb 21 17:19 Library
drwxr-xr-x 6 kali kali 4096 Feb 21 17:19 private
drwxr-xr-x 2 kali kali 4096 Feb 21 17:19 sbin
drwxr-xr-x 3 kali kali 4096 Feb 21 17:18 System
lrwxrwxrwx 1 kali kali 15 Apr 2  2020 tmp -> private/var/tmp
drwxr-xr-x 10 kali kali 4096 Feb 21 17:18 usr
lrwxrwxrwx 1 kali kali 11 Apr 2  2020 var -> private/var
```

Root Level Folder View:



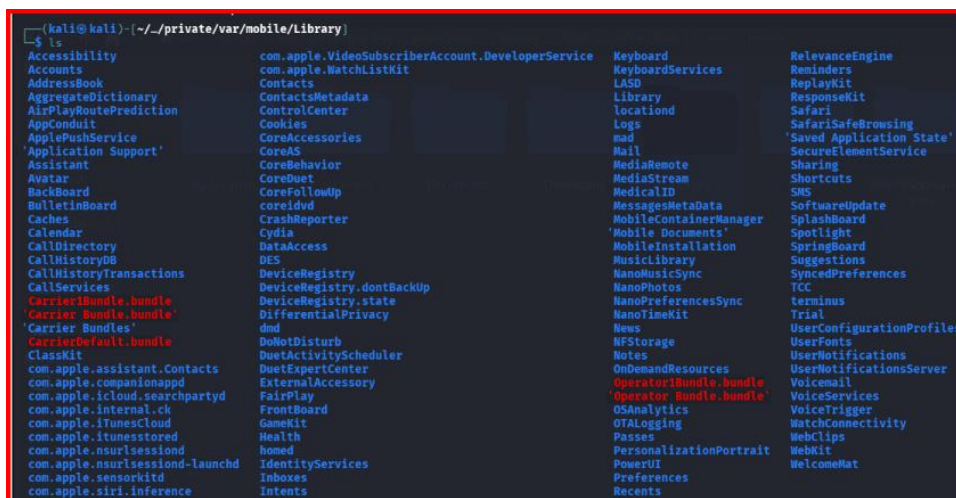
~/System/Library: iOS Building Block



Where are applications installed?

Application data is stored in: `~/var/containers/Bundle/Application`- This contains code, apple signatures, and frameworks, no user data.

Applications user data is stored in: `~/var/mobile/Containers/Data/Application`- This contains user generated data and preferences

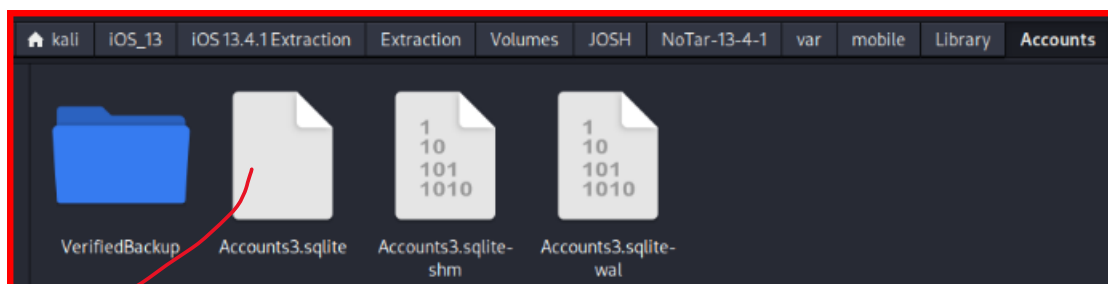


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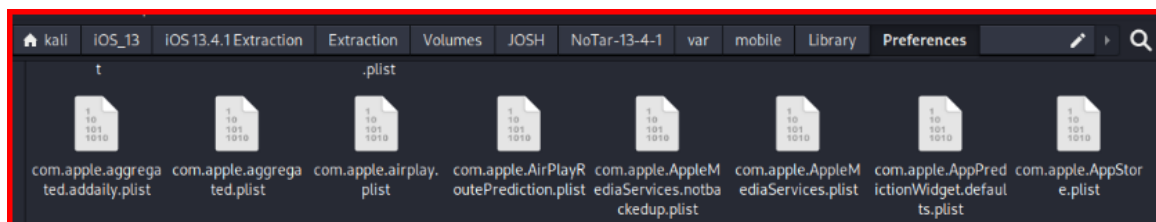
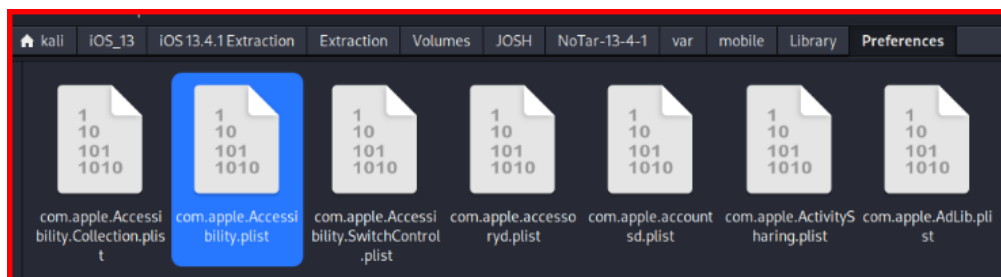
I Favour Madu confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.


```
(kali@kali) [~/var/containers/Bundle/Application]
$ tree -L 2
.
├── 00DF1C62-AA0F-4391-91F6-321C9A1A41A0
│   ├── BundleMetadata.plist
│   └── iTunesMetadata.plist
├── 036E8380-8011-4109-9315-8919F5C3E39C
│   ├── BundleMetadata.plist
│   └── iTunesMetadata.plist
├── 04DA2653-4349-45BA-8F3C-760556CFAD02
│   └── BundleMetadata.plist
├── 06703F45-064C-4031-A617-EEB4882C036
│   ├── BundleMetadata.plist
│   └── iTunesMetadata.plist
├── 07FFD43B-C536-499A-876E-CE78DB853061
│   ├── BundleMetadata.plist
│   └── iTunesMetadata.plist
├── 08552D76-A024-447A-863A-5E77D3F7965D
│   ├── BundleMetadata.plist
│   └── iTunesMetadata.plist
├── 0944046B-B60A-4FDD-80FA-F0A85045CB4E
│   ├── BundleMetadata.plist
│   └── iTunesMetadata.plist
├── 0F406F07-1C32-43B7-A850-D9DBDCAD51DE
│   ├── BundleMetadata.plist
│   └── iTunesMetadata.plist
├── 19C77A91-D4D9-488C-AC0F-F613693D4F41
│   └── BundleMetadata.plist
└── iTunesMetadata.plist
```

Generally Important .plists & .dbs

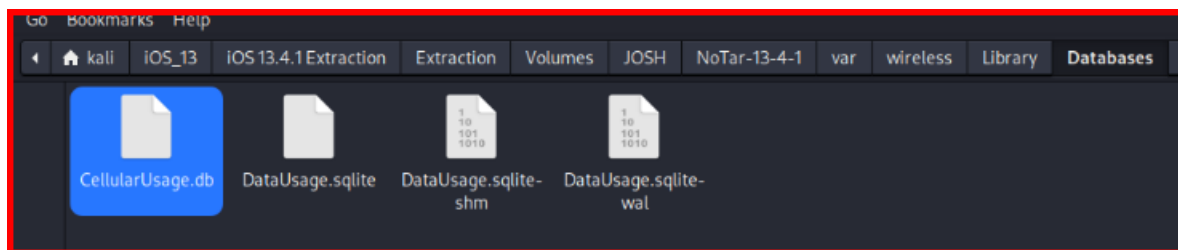


Showing error message when trying to view on sqli browser



Academic Integrity and Authorization Statement

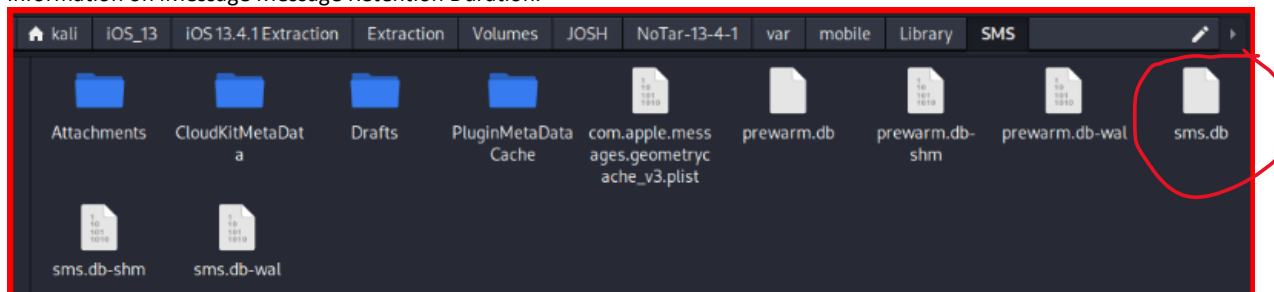
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Sms.db – Found at ~/private/var/mobile/Library/SMS

Contains:

Information on iMessage Message Retention Duration.



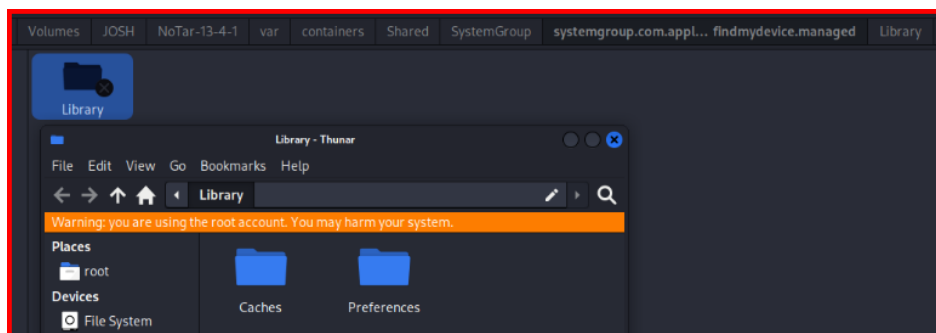
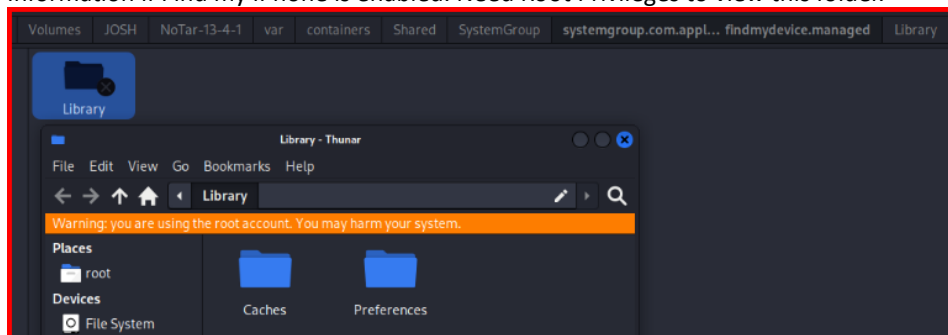
Generally Important .plist & .dbs (5)

Com.apple.icloud.findmydevice.FMIPAccounts.plist – Found at

~/private/var/containers/Shared/SystemGroup/systemgroup.com.apple.icloud.findmydevice.managed/Library/Preferences

Contains:

Information if Find my iPhone is enabled. Need Root Privileges to view this folder.

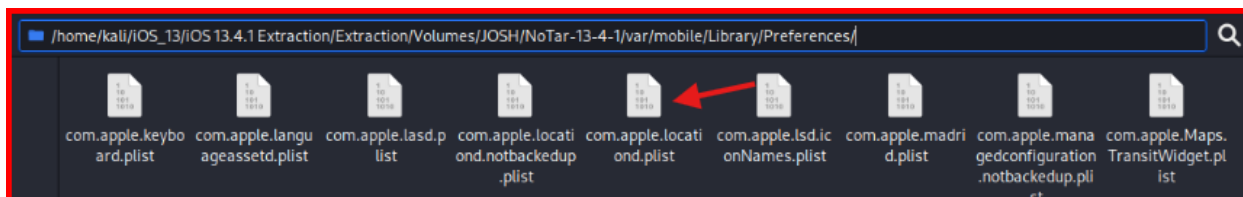


Com.apple.locationd.plist - Found at ~/private/var/mobile/Library/Preferences

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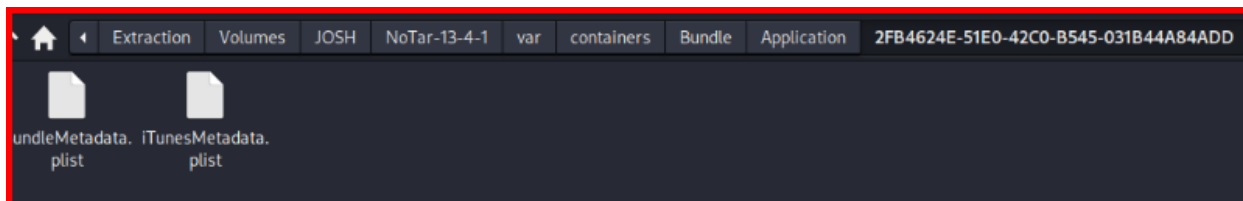
Contains: Information if Location services are enables.



Understanding Application Bundles

Table 2-2 Contents of a typical iOS application bundle

File	Description
MyApp	(Required) The executable file containing your application's code. The name of this file is the same as your application name minus the <code>.app</code> extension.
Application icons (MyAppIcon.png, MySearchIcon.png, and MySettingsIcon.png)	(Required/Recommended) Application icons are used at specific times to represent the application. For example, different sizes of the application icon are displayed in the Home screen, in search results, and in the Settings application. Not all of the icons are required but most are recommended. For information about application icons, see Application Icon and Launch Images .
Info.plist	(Required) This file contains configuration information for the application, such as its bundle ID, version number, and display name. See The Information Property List File for further information.
Launch images (Default.png)	(Recommended) One or more images that show the initial interface of your application in a specific orientation. The system uses one of the provided launch images as a temporary background until your application loads its window and user interface. If your application does not provide any launch images, a black background is displayed while the application launches. For information about application icons, see Application Icon and Launch Images .
MainWindow.nib	(Recommended) The application's main nib file contains the default interface objects to load at application launch time. Typically, this nib file contains the application's main window object and an instance of the application delegate object . Other interface objects are then either loaded from additional nib files or created programmatically by the application. (The name of the main nib file can be changed by assigning a different value to the <code>NSMainNibFile</code> key in the <code>Info.plist</code> file. See The Information Property List File for further information.)
Settings.bundle	The Settings bundle is a special type of plug-in that contains any application-specific preferences that you want to add to the Settings application. This bundle contains property lists and other resource files to configure and display your preferences.
Custom resource files	Non-localized resources are placed at the top level directory and localized resources are placed in language-specific subdirectories of the application bundle. Resources consist of nib files , images, sound files, configuration files, strings files, and any other custom data files you need for your application. For more information about resources, see Resources in an iOS Application .



iPhone iOS Device Investigations

General Device Information, Account Information, Jailbreak

cd into `/var/installd/Library/MobileInstallation`

`plutil -i LastBuildInfo.plist`

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```
(kali@kali)~[~/var/install/Library/MobileInstallation]
$ plistutil -i LastBuildInfo.plist

<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
  <key>SystemImageID</key>
  <string>747CABD1-394F-4F90-92E9-CC6B5E1699DE</string>
  <key>Version</key>
  <string>Version</string>
  <key>ProductVersion</key>
  <string>13.4.1</string>
  <key>BuildID</key>
  <string>9695BE08-7547-11EA-B2DF-F0563A0B6C5E</string>
  <key>ProductBuildVersion</key>
  <string>17E262</string>
  <key>ProductCopyright</key>
  <string>1983-2020 Apple Inc.</string>
  <key>Build</key>
  <string>Build</string>
  <key>ProductName</key>
  <string>iPhone OS</string>
  <key>FullVersionString</key>
  <string>Version 13.4.1 (Build 17E262)</string>
</dict>
</plist>
```

What is the phone number?

~/private/var/wireless/Library/Preferences

plistutil -i com.apple.commcenter.plist

```
(kali@kali)~[~/var/wireless/Library/Preferences]
$ plistutil -i com.apple.commcenter.plist

<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
  <key>SIMPhoneNumber</key>
  <string>+19195794674</string>
  <key>PersonalitySlots</key>
  <array>
    <string>8901260971148676693</string>
  </array>
</dict>
```

The phone number registered to the SIM card inside this iPhone is **+19195794674**

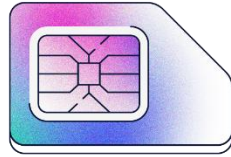
What is the device's ICCID?

<string>8901260971148676693</string>

ICCID: 8901260971148676693

What is the device's IMSI?

```
</dict>
<key>CarrierEntitlements</key>
<dict>
  <key>lastGoodImsi</key>
  <string>310260974867669</string> IMSI
  <key>kEntitlementsSelfRegistrationUpdateImsi</key>
  <string>310260974867669</string>
  <key>kEntitlementsSelfRegistrationUpdateImsi</key>
  <string>355800076093966</string>
  <key>phone-number-prefetched</key>
  <true/>
```



Components of IMSI

3 1 0 4 1 0 2 3 4 5 6 2 3 9

MCC	MNC	Remaining Digits
• Mobile Country Code • First 2-3 digits	• Mobile Network Code • Next 2-3 digits	• Line of service • Also called subscriber

What are the device's MCC & MNC codes?

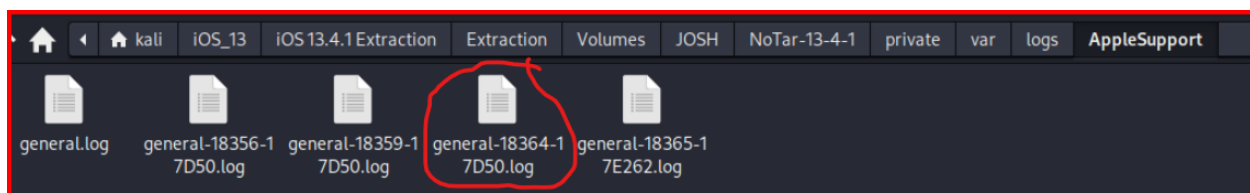
Mobile Country Code (MCC) and Mobile Network Code (MNC) are used to find & identify network providers.

```
<key>LastKnownServingMnc</key>
  <integer>260</integer> MNC: 260 (T-Mobile
  <key>PhoneNumberChangeReport</key>
  <true/>
<key>LastKnownServingMcc</key>
  <integer>310</integer> MCC: 310 (US
  <key>NetworkPhoneNumber</key>
```

What is the serial number?

```
(kali@kali)-[~/private/var/logs/AppleSupport]
$ ls
general-18356-17D50.log general-18359-17D50.log general-18364-17D50.log general-18365-17E262.log general.log
(kali@kali)-[~/private/var/logs/AppleSupport]
$ cat general.log
{"report": "Device Software Diagnostic Log", "serial": "DX3T126VH2XV", "os_build": "iPhone OS 13.4.1 (17E262)", "updated": "2020-04-13 15:15:32 -0400", "model": "iPhone8,4", "version": "4", "installed": "2020-03-21 17:39:05 -0400"}
```

What is the device's model number?



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```

~/.iOS_13/iOS 13.4.1 Extraction/Extraction/Volumes/JOSH/NoTar-13-4-1/private/var/logs/AppleSu
File Edit Search View Document Help
1 [{"report": "Device Software Diagnostic
Log", "serial": "DX3T126VH2XV", "os_build": "iPhone OS 13.3.1
(17D50)", "updated": "2020-03-21 17:39:05
-0400", "model": "iPhone8,4", "version": 4, "installed": "2020-03-21 17:39:05
-0400"}]
2

```

iPhone8

Account information

DB Browser for SQLite - /home/kali/iOS_13/iOS 13.4.1 Extraction/Extraction/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Accounts/Accounts3.sqlite

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database

Database Structure Browse Data Edit Pragmas Execute SQL

Create Table Create Index Modify Table Delete Table Print Refresh

Name	Type	Schema
Tables (15)		
ZACCESSOPTIONSKEY	Browse Table	SOPTIONSKEY (Z_PK INTEGER PRIMARY KEY, Z_ENT INTI
ZACCOUNT	Modify Table	JNT (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OI
ZACCOUNTPROPERTY	Delete Table	JNTPROPERTY (Z_PK INTEGER PRIMARY KEY, Z_ENT INTI
ZACCOUNTTYPE	Copy Create statement	JNTTYPE (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER,
ZAUTHORIZATION	Export as CSV file	ORIZATION (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGE
ZCREDENTIALITEM		INITIALITEM (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGE
ZDATACLASS		:LASS (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_C
Z_IOWNINGACCOUNTTYPES		CREATE TABLE Z_IOWNINGACCOUNTTYPES (Z_1ACCESSKEYS INTEGER, Z_4OW
Z_2ENABLEDDATACLASSES		CREATE TABLE Z_2ENABLEDDATACLASSES (Z_2ENABLEDDATAACCOUNTS INTEGER, ;
Z_2PROVISIONEDDATACLASSES		CREATE TABLE Z_2PROVISIONEDDATACLASSES (Z_2PROVISIONEDDATAACCOUNTS II
Z_4SUPPORTEDDATACLASSES		CREATE TABLE Z_4SUPPORTEDDATACLASSES (Z_4SUPPORTEDTYPES INTEGER,
Z_4SYNCABLEDATACLASSES		CREATE TABLE Z_4SYNCABLEDATACLASSES (Z_4SYNCABLETYPES INTEGER, Z_;
Z_METADATA		CREATE TABLE Z_METADATA (Z_VERSION INTEGER PRIMARY KEY, Z_UUID VARCH
Z_MODELCACHE		CREATE TABLE Z_MODELCACHE (Z_CONTENT BLOB)
Z_PRIMARYKEY		CREATE TABLE Z_PRIMARYKEY (Z_ENT INTEGER PRIMARY KEY, Z_NAME VARCHA
Indices (9)		
ZACCOUNTPROPERTY_ZOWNER_IN...		CREATE INDEX ZACCOUNTPROPERTY_ZOWNER_INDEX ON ZACCOUNTPROPER
ZACCOUNT_ZACCOUNTTYPE_INDEX		CREATE INDEX ZACCOUNT_ZACCOUNTTYPE_INDEX ON ZACCOUNT (ZACCOUNT
ZACCOUNT_ZPARENTACCOUNT_IN...		CREATE INDEX ZACCOUNT_ZPARENTACCOUNT_INDEX ON ZACCOUNT (ZPAREN
ZAUTHORIZATION_ZACCOUNTTYPE...		CREATE INDEX ZAUTHORIZATION_ZACCOUNTTYPE_INDEX ON ZAUTHORIZATIC
Z_IOWNINGACCOUNTTYPES_Z_4O...		CREATE INDEX Z_IOWNINGACCOUNTTYPES_Z_4OWNINGACCOUNTTYPES_IND
Z_2ENABLEDDATACLASSES_Z_7ENA...		CREATE INDEX Z_2ENABLEDDATACLASSES_Z_7ENABLEDDATACLASSES_INDEX I
Z_2PROVISIONEDDATACLASSES_Z_7...		CREATE INDEX Z_2PROVISIONEDDATACLASSES_Z_7PROVISIONEDDATACLASSE
Z_4SUPPORTEDDATACLASSES_Z_7S...		CREATE INDEX Z_4SUPPORTEDDATACLASSES_Z_7SUPPORTEDDATACLASSES_IN
Z_4SYNCABLEDATACLASSES_Z_7SY...		CREATE INDEX Z_4SYNCABLEDATACLASSES_Z_7SYNCABLEDATACLASSES_INDE
Views (0)		
Triggers (0)		

Edit Database Cell

Mode: Text

1

ZACCOUNT									
Z_PK	Z_ENT	Z_OPT	ZACTIVE	ZAUTHENTICATED	ZSUPPORTSAUTHENTICATION	ZVISIBLE	ZACCOUNTTYPE	ZPARENTACC	
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	
1	1	2	8	0	1	1	25		
2	2	2	60	1	1	1	25		
3	3	2	77	1	1	1	40		
4	4	2	43	1	1	1	10		
5	5	2	15	1	1	0	24		
6	6	2	51	1	1	0	19		

Academic Integrity and Authorization Statement

I Favour Madu confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.

```
(kali@kali)-[~/../var/mobile/Library/Preferences]
$ plistutil -i com.apple.mobile.ldbbackup.plist
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
  <dict>
    <key>CloudBackupEnabled</key>
    <true/>
    <key>Version</key>
    <string>2.0</string>
    <key>LastiTunesBackupTZ</key>
    <string>EDT</string>
    <key>LastiTunesBackupDate</key>
    <integer>608744930</integer>
    <key>LastCloudBackupTZ</key>
    <string>EDT</string>
    <key>WillEncrypt</key>
    <true/>
    <key>RequiresEncryption</key>
    <integer>0</integer>
    <key>LastCloudBackupDate</key>
    <integer>608693130</integer>
  </dict>
</plist>
```

Location Services

```
(kali@kali)-[~/../var/mobile/Library/Preferences]
$ plistutil -i com.apple.locationd.plist
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
  <dict>
    <key>LastSystemVersion</key>
    <string>iPhone OS13.4.1/17E262</string>
    <key>MotionFitnessDailyToken</key>
    <string>A13F0FBE-92A7-43CD-B4CF-4D4A92D9C148</string>
    <key>kCLVersionOfLastFullSync</key>
    <integer>4</integer>
    <key>CLSilos.Version</key>
    <integer>1</integer>
    <key>ObsoleteDataDeleted</key>
    <true/>
    <key>OscarStepCounting</key>
    <true/>
    <key>LocationServicesEnabledIn8.0</key>
    <true/>
  </dict>
</plist>
```

Jailbreak

The term jailbreak is the process by which full execute and write access is obtained on all the partitions of iOS. Modern jailbreaks include patching the kernel to get around code signing and other restrictions.

There are two types of jailbreak semi-tethered and semi-untethered.

When a device begins booting, it loads Apple's own kernel initially, so a jailbroken device must be exploited and have the kernel patched each time it is booted up

Semi-tethered solutions are the least popular form of jailbreaking. Every time the user reboots their phones the jailbreak will be removed, meaning they must tether their phone to their computer and run the jailbreak software to keep the jailbroken state on their device.

Semi-untethered is the most popular form of jailbreak which allows users to reboot their phones and resume the jailbreaking features without having to attach to a computer to patch the kernel.

Finding evidence of Jailbreak

```
(kali@kali)-[~/JOSH/NoTar-13-4-1/private/var]
$ ls -l
total 7008
drwx----- 2 kali kali 4096 Apr 13 2020 audit
drwxr-xr-x 2 kali kali 4096 Oct 28 2006 backups
drwxr-xr-x 2 kali kali 4096 Feb 21 17:19 binpack
drwx----- 2 kali kali 4096 Jan 18 2020 buddy
drwxr-xr-x 2 kali kali 4096 Apr 10 2019 cache
-rw-r--r-- 1 kali kali 7025305 Apr 12 2020 checkra1n.dmg
drwxr-xr-x 5 kali kali 4096 Feb 21 17:19 containers
drwxr-xr-x 25 kali kali 4096 Feb 21 17:19 db
-rw----- 1 kali kali 805 Apr 12 2020 dropbear_rsa_host_key
drwxr-xr-x 2 kali kali 4096 Jan 18 2020 empty
drwxr-xr-x 2 kali kali 4096 Jan 18 2020 folders
drwxr-xr-x 2 kali kali 4096 Apr 13 2020 hardware
drwxr-xr-x 3 kali kali 4096 Feb 21 17:19 install
drwxr-xr-x 2 kali kali 4096 Mar 21 2020 iomfb_bics_daemon
drwx----- 3 kali kali 4096 Feb 21 17:19 keybags
drwxr-xr-x 5 kali kali 4096 Feb 21 17:19 Keychains
drwxr-xr-x 5 kali kali 4096 Feb 21 17:19 lib
drwxrwxr-x 2 kali kali 4096 Oct 28 2006 local
drwxrwxr-x 2 kali kali 4096 Oct 28 2006 lock
drwxr-xr-x 9 kali kali 4096 Feb 21 17:19 log
drwxr-xr-x 7 kali kali 4096 Feb 21 17:19 logs
drwxr-xr-x 3 kali kali 4096 Feb 21 17:19 'Managed Preferences'
drwx--x--x 9 kali kali 4096 Feb 21 17:19 mobile
drwxr-xr-x 5 kali kali 4096 Feb 21 17:19 MobileAsset
drwxr-xr-x 3 kali kali 4096 Feb 21 17:19 MobileDevice
drwxr-xr-x 9 kali kali 4096 Feb 21 17:19 MobileSoftwareUpdate
drwxr-xr-x 2 kali kali 4096 Jan 18 2020 msgs
drwxr-xr-x 3 kali kali 4096 Feb 21 17:19 networkd
drwxr-xr-x 4 kali kali 4096 Feb 21 17:19 preferences
drwxr-xr-x 4 kali kali 4096 Feb 21 17:19 root
drwxrwxr-x 2 kali kali 4096 Feb 21 17:19 run
drwxr-xr-x 2 kali kali 4096 Apr 13 2020 select
drwxr-xr-x 2 kali kali 4096 Oct 28 2006 spool
drwxrwxr-x 2 kali kali 4096 Apr 13 2020 staged_system_apps
drwxrwxr-x 39 kali kali 4096 Feb 21 17:19 tmp
drwxr-xr-x 2 kali kali 4096 Feb 21 17:19 vm
drwxr-xr-x 6 kali kali 4096 Mar 21 2020 wireless
```

Another place to check is `~/var/Mobile/Library`. Specially looking for a Cydia folder.

Cydia is an open-source software installer that uses the Debian APT system for package management.

```
(kali@kali)-[~/private/var/mobile/Library]
$ ls
Accessibility com.apple.VideoSubscriberAccount.DeveloperService Keyboard RelevanceEngine
Accounts com.apple.WatchListKit KeyboardServices Reminders
AddressBook Contacts LASD ReplayKit
AggregateDictionary ContactsMetadata Library ResponseKit
AirPlayRoutePrediction ControlCenter locationd Safari
AppConduit Cookies Logs SafariSafeBrowsing
ApplePushService CoreAccessories mail 'Saved Application State'
'Application Support' CoreAS Mail SecureElementService
Assistant CoreBehavior MediaRemote Sharing
Avatar CoreDuet MediaStream Shortcuts
BackBoard CoreFollowUp MedicalID SMS
BulletinBoard CoreData MessagesMetadata SoftwareUpdate
Caches CrashReporter MobileContainerManager SplashBoard
Calendar cydia MobileDocuments Spotlight
CallDirectory DataAccess MobileInstallation SpringBoard
CallHistoryDB DES MusicLibrary Suggestions
CallHistoryTransactions DeviceRegistry NanoMusicSync SyncedPreferences
CallServices DeviceRegistry.dontBackup TCC
CarrierBundle.bundle DeviceRegistry.state terminus
'Carrier Bundle.bundle' DifferentialPrivacy NanoTimeKit Trial
'Carrier Bundles' dmd News NFStorage UserConfigurationProfiles
CarrierDefault.bundle DoNotDisturb Notes OnDemandResources UserNotifications
ClassKit DuetActivityScheduler OperatorBundle.bundle Voicemail
com.apple.assistant.contacts DuetExpertCenter 'Operator Bundle.bundle' VoiceServices
com.apple.companionappd ExternalAccessory OSAnalytics VoiceTrigger
com.apple.icloud.searchpartyd FairPlay OTALogging WatchConnectivity
com.apple.internal.ck FrontBoard Passes WebClips
com.apple.iTunesCloud GameKit PersonalizationPortrait WebKit
com.apple.itunesstored Health Preferences WelcomeMat
com.apple.nsurlsessiond homed PowerUI
com.apple.nsurlsessiond-launchd IdentityServices Recents
com.apple.sensorkitd Inboxes
com.apple.siri.inference Intents
```

Academic Integrity and Authorization Statement

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iOS System and App Settings

Where are settings stored from the Settings App?

```
(kali@kali)-[~/mobile/Library/UserConfigurationProfiles/PublicInfo]
$ ls
MCMeta.plist  NamespacedUserSettings.plist  PublicEffectiveUserSettings.plist  Truth.plist
```

Examining System Settings

```
(kali@kali)-[~/mobile/Library/UserConfigurationProfiles/PublicInfo]
$ plistutil -i PublicEffectiveUserSettings.plist

<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
  <key>union</key>
  <dict>
    <key>audioCallDefaultHandlerAppBundleIDs</key>
    <dict/>
    <key>knownAirPrintTargets</key>
    <dict/>
    <key>tvRemoteAllowedRemotes</key>
    <dict/>
    <key>managedWebDomains</key>
    <dict/>
    <key>trustedCodeSigningIdentities</key>
    <dict/>
    <key>removedSystemAppBundleIDs</key>
    <dict>
      <key>values</key>
      <array>
        <string>com.apple.shortcuts</string>
        <string>com.apple.iBooks</string>
        <string>com.apple.tv</string>
        <string>com.apple.tips</string>
        <string>com.apple.iCloudDriveApp</string>
      </array>
    </dict>
    <key>tvRemoteAllowedTVs</key>
    <dict/>
    <key>allowedSafariPasswordAutoFillDomains</key>
    <dict/>
    <key>webContentFilterBlacklistedURLs</key>
    <dict/>
  </dict>
</dict>
```

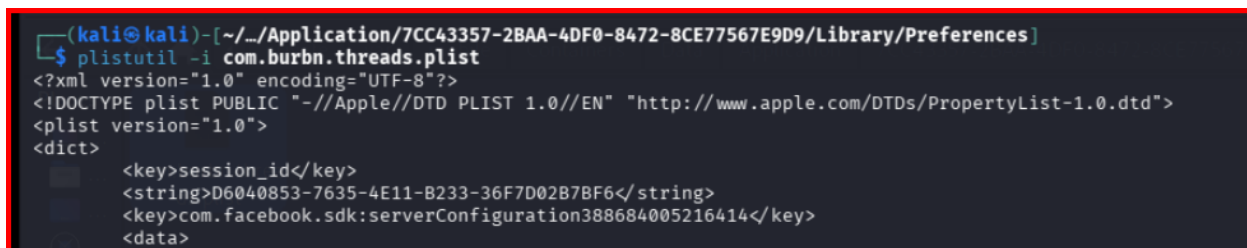
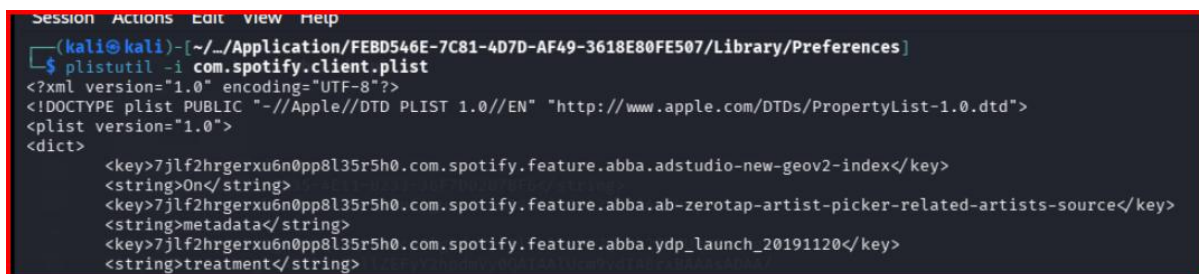
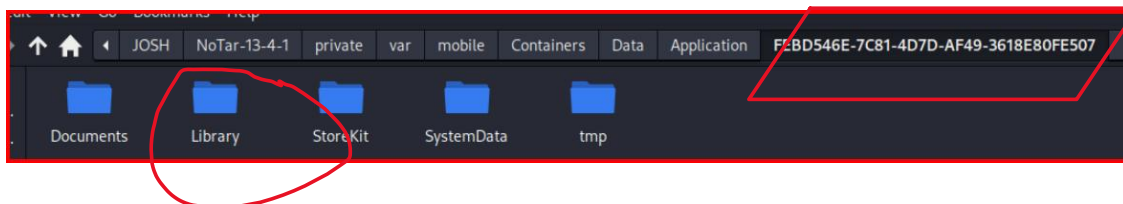
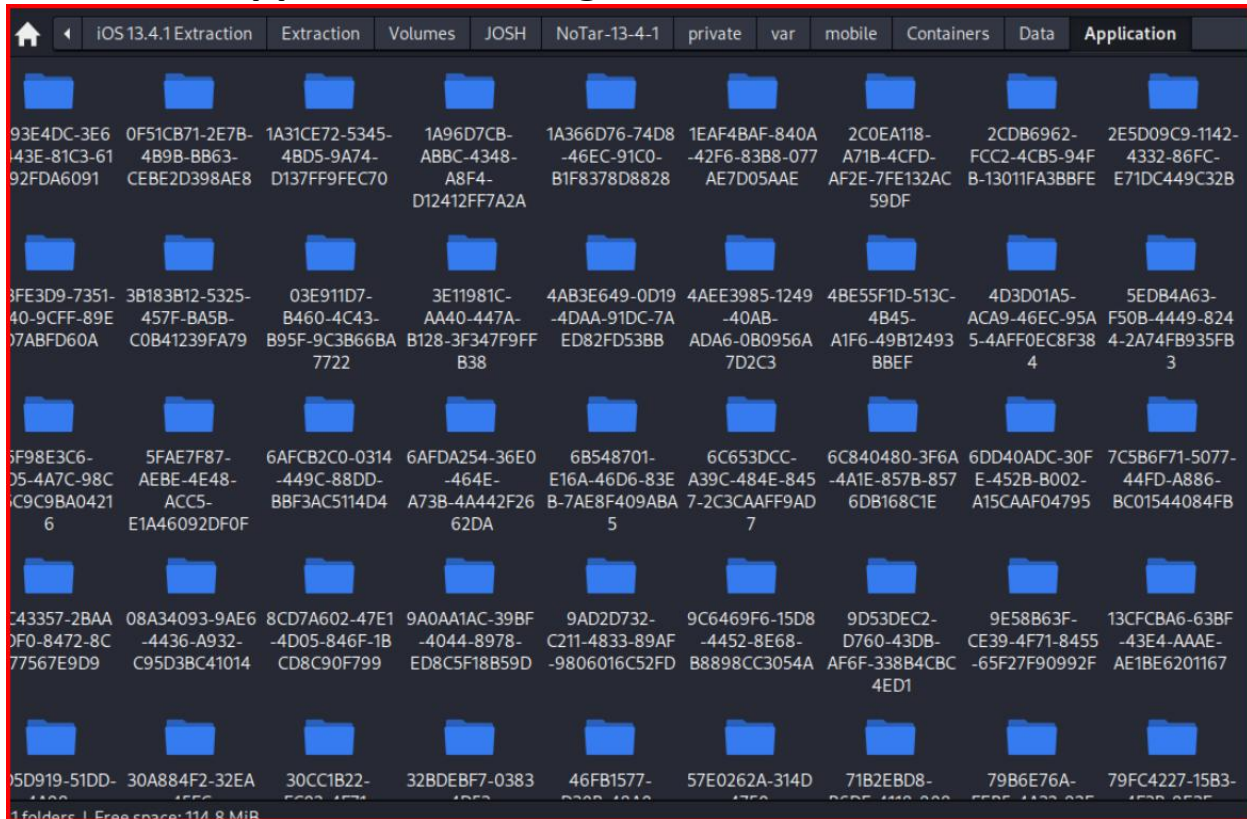
Examples from XML 1

```
</dict>
<key>forceAutomaticDateAndTime</key>
<dict>
  <key>value</key>
  <false/>
</dict>
```

2

```
</dict>
<key>allowControlCenterInApps</key>
<dict>
  <key>value</key>
  <true/>
</dict>
```


Where are app related settings stored on iOS



Academic Integrity and Authorization Statement

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iOS Messages Investigation

- Introduction to Messages App
- Investigating Messaging Database

SMS (short message service)

a text messaging service component of most telephone, Internet, and mobile device systems

MMS (Multimedia Message Service)

send multimedia content—pictures, video, and audio—in addition to text

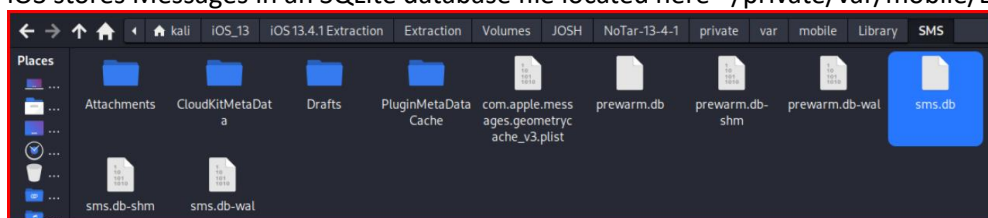
What is iMessage?

Messages is the default texting app for iOS on any iPhone, iPod touch, and iPad. It lets you do all the basic things you'd expect: Send texts, photos, emojis, and all the other standard texting stuff.

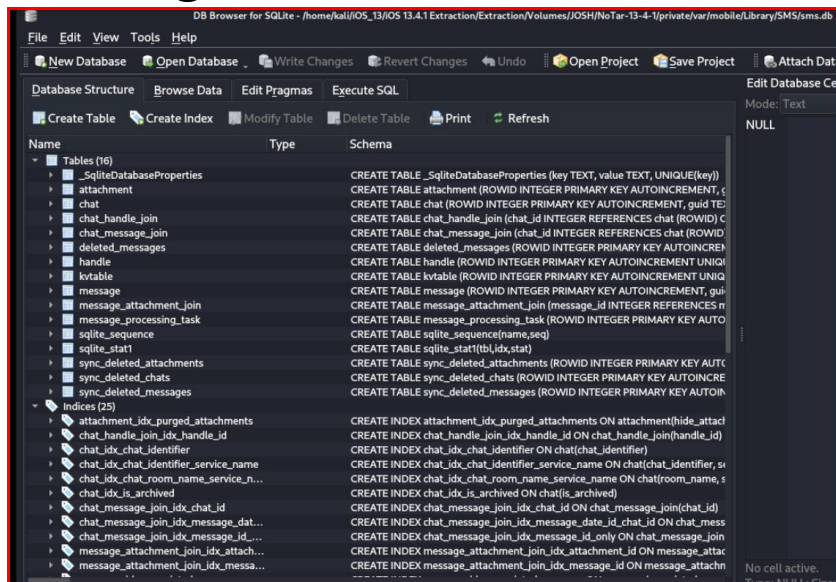
On the other hand, iMessage is an Apple-specific set of features and tools that are built on top of Messages. The advanced features used in Messages come from iMessage. You can use other apps to send texts from your iPhone, but if you want to use all iMessage's features, your only option is to use the Messages app.

Investigating Messages Database

iOS stores Messages in an SQLite database file located here ~/private/var/mobile/Library/SMS/sms.db



Viewing sms.db



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Attachment Table

_SqliteDatabaseProperties

attachment

attachment

Table: attachment

Filter in any column

Wl	guid	created_date	start_date	filename
...	Filter	Filter	Filter	Filter
1	1 at_0_57633602-7DC8-4506-AA44-9F06D82CA654	607027332	0	~/Library/SMS/Attachments/cb/11/F1E68FA1-1F02-4544-A1C2-E9FC69565B4D/AF54E2D6-D6C5-4652-...
2	2 6B32D057-049C-405B-8A4A-CAE8B5C201E1	608389002	0	/var/tmp/com.apple.messages/com.apple.MobileSMS/Media/E54EFEAC-40A3-45D3-88BD-84B3E0A576FC...
3	3 at_0_784979CA-1E85-4FFE-A2CE-1AD0B388209E	608389004	0	~/Library/SMS/Attachments/10/00/A0159B39-9B33-401F-B5F2-876388BCF97B/IMG_0008.JPG
4	4 at_0_c8BCB97D-9503-48BD-842D-7618ADD1631E	608389082	608389082	~/Library/SMS/Attachments/54/04/9A5EF4E2-87A7-4923-ABD7-218D94989A09/FullSizeRender.jpeg
5	5 at_0_D3C8A063-6CEB-4326-84BB-10D68D9F2E50	608390156	0	~/Library/SMS/Attachments/68/08/430548CB-3157-418F-9B0E-3E677784E7A4/How was an iPhone set...
6	6 at_0_6A995D6E-14E2-4620-8826-8F4EE2926FE5	608390475	608390476	~/Library/SMS/Attachments/97/07/476F1BF4-4FE5-466A-AD8C-605576C5F16D/...
7	7 at_0_E1DE4D6D-5E22-44CB-874A-A173AAA77DE0	608730349	0	~/Library/SMS/Attachments/db/11/DDDBC8B0-6279-49B0-85B1-ADC00A191D32/069EBD0E-FDA6-486A-...

```
echo $((978307200+607027332))
```

```
date -d @1585334532
```

<https://www.epochconverter.com/coredata>

```
(kali@kali)-[~/../var/mobile/Library/SMS]
$ echo $((978307200+607027332))
1585334532

(kali@kali)-[~/../var/mobile/Library/SMS]
$ date -d @1585334532 -u
Sun May 15 07:02:02 PM UTC 2472

(kali@kali)-[~/../var/mobile/Library/SMS]
$ date -d @1585334532
Sun May 15 03:02:02 PM EDT 2472
```

attachment											
Table: attachment											
	mime_type	transfer_state	is_outgoing	user_info	transfer_name	total_bytes	is_sticker	sticker_user_info	attribution_info	hide_attachment	ck_syr
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	NULL	5	1	NULL	AF54E2D6-D6C5-4652-...	770534	0	NULL	NULL	1	
2	image/jpeg	0	1	NULL	IMG_0008.JPG	676542	0	NULL	NULL	0	
3	image/jpeg	5	1	BLOB	IMG_0008.JPG	676542	0	NULL	BLOB	0	
4	image/jpeg	5	0	BLOB	FullSizeRender.jpeg	368536	0	NULL	BLOB	0	
5	application/pdf	5	1	BLOB	How was an iPhone set up?.pdf	846528	0	NULL	NULL	0	
6	application/pdf	5	0	BLOB	iOS_Bug_Reporting_for_Forensic_Purpo...	7073725	0	NULL	NULL	0	
7	NULL	5	1	NULL	069EBD0E-FDA6-486A-...	1598	0	NULL	NULL	1	

CHAT TABLE

Academic Integrity and Authorization Statement

I Favour Madu confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.

chat

Table: chat

Filter in any column

ROWID	guid	style	state	account_id	properties	chat_idenfier	service_name	room_name	account_
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1 SMS; -; 22000	45	3	E087E708-B720-4183-AB43-90A9CC7FDC61	BLOB	22000	SMS	NULL	P: +1919579467
2	2 SMS; -; +18634205597	45	3	E087E708-B720-4183-AB43-90A9CC7FDC61	BLOB	+18634205597	SMS	NULL	P: +1919579467
3	3 SMS; -; +13176444906	45	3	E087E708-B720-4183-AB43-90A9CC7FDC61	BLOB	+13176444906	SMS	NULL	P: +1919579467
4	4 SMS; -; 85760	45	3	E087E708-B720-4183-AB43-90A9CC7FDC61	BLOB	85760	SMS	NULL	P: +1919579467
5	5 SMS; -; +17026604496	45	3	E087E708-B720-4183-AB43-90A9CC7FDC61	BLOB	+17026604496	SMS	NULL	P: +1919579467
6	6 SMS; -; 32665	45	3	E087E708-B720-4183-AB43-90A9CC7FDC61	BLOB	32665	SMS	NULL	P: +1919579467
7	7 SMS; -; +12074242620	45	3	E087E708-B720-4183-AB43-90A9CC7FDC61	BLOB	+12074242620	SMS	NULL	P: +1919579467
8	8 SMS; -; +13233208923	45	3	E087E708-B720-4183-AB43-90A9CC7FDC61	BLOB	+13233208923	SMS	NULL	P: +1919579467
9	9 SMS; -; 47543	45	3	E087E708-B720-4183-AB43-90A9CC7FDC61	BLOB	47543	SMS	NULL	P: +1919579467
10	10 SMS; -; +12349008110	45	3	E087E708-B720-4183-AB43-90A9CC7FDC61	BLOB	+12349008110	SMS	NULL	P: +1919579467
11	11 SMS; -; 732873	45	3	E087E708-B720-4183-AB43-90A9CC7FDC61	BLOB	732873	SMS	NULL	P: +1919579467
12	12 SMS; -; +13214065930	45	3	E087E708-B720-4183-AB43-90A9CC7FDC61	BLOB	+13214065930	SMS	NULL	P: +1919579467
13	13 iMessage; -; joshua.hickman1@me.com	45	3	C16C919A-EF22-4C56-AA34-68196C598773	BLOB	joshua.hickman1@me.com	iMessage	NULL	E: thisisdfr
14	14 SMS; -; +18334030436	45	3	E087E708-B720-4183-AB43-90A9CC7FDC61	BLOB	+18334030436	SMS	NULL	P: +1919579467
15	15 SMS; -; 78156	45	3	E087E708-B720-4183-AB43-90A9CC7FDC61	BLOB	78156	SMS	NULL	P: +1919579467
16	16 SMS; -; 4439A	45	3	F087F708-B720-4183-AB43-90A9CC7FDC61	BLOB	4439A	SMS	NULL	P: +1919579467

Go to: 1

Message Attachment

_SqliteDatabaseProperties message

message

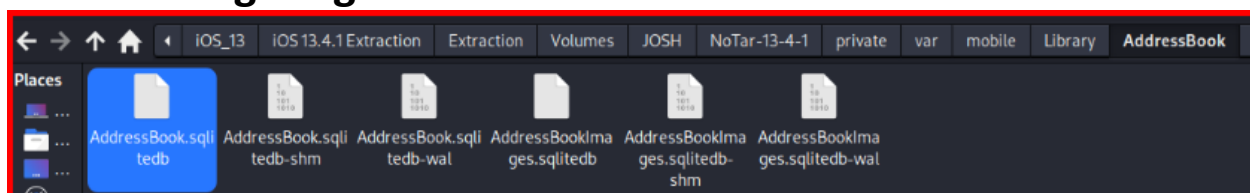
Table: message

Filter in any column

ROWID	guid	text	replace	service_center	handle_id	subject	country	attributedBo
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	6638E204-1917-C74D-68F5-7925D8AAD333	G-773293 is your Google verification...	0	NULL	1	NULL	NULL	BLOB
2	AF1E7300-055D-F2CA-3969-416C8880A19A	imo code: 258465...	0	NULL	2	NULL	NULL	BLOB
3	6DA20C06-0BAE-1609-FE7F-7054BF58367A	Please enter 737950 into LINE within...	0	NULL	3	NULL	NULL	BLOB
4	8B454216-2683-8567-C6EB-E471104D22B1	Please enter this code in the app to...	0	NULL	4	NULL	NULL	BLOB
5	0ACC60B1-1FEB-6038-CC1C-45B7ABDF9F03	Your Signal verification code: ...	0	NULL	5	NULL	NULL	BLOB
6	F02C44BF-499E-7FEA-DC13-1C1993676D16	🔔 This is, you have 2 new ...	0	NULL	6	NULL	NULL	BLOB
7	B0A4B871-4EA8-ADCF-C03E-E6E6B5444EAC	Telegram code: 95037...	0	NULL	7	NULL	NULL	BLOB
8	2A1B26B6-DE6E-E52C-44EA-FFE86D108C32	[TikTok] 8494 is your verification ...	0	NULL	8	NULL	NULL	BLOB

Investigating Contacts, Voicemails, and Call Logs

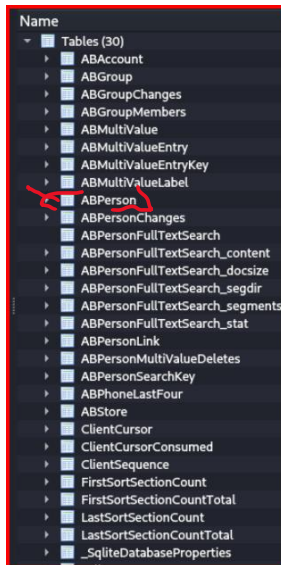
Investigating Contacts



Examining the ABPerson table (2)

Academic Integrity and Authorization Statement

I **Favour Madu** confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.



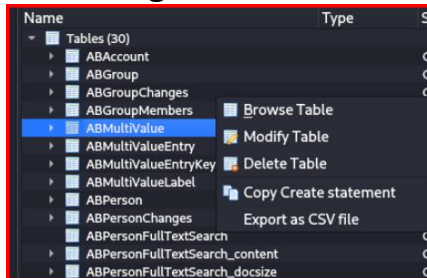
ABAccount ABPerson

ABPerson

Table: ABPerson

ROWID	First	Last	Middle	FirstPhonetic	MiddlePhonetic	LastPhonetic	Organization	Department	Note	Kind	Birthday	JobTitle	Nickname	Prefix	S
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	Josh	Hickman	NULL	NULL	NULL	NULL	NULL	NULL	0	NULL	NULL	NULL	NULL	N
2	2	This	Is	DFIR	NULL	NULL	NULL	NULL	NULL	0	NULL	NULL	NULL	NULL	N

Examining the ABMultiValue Table



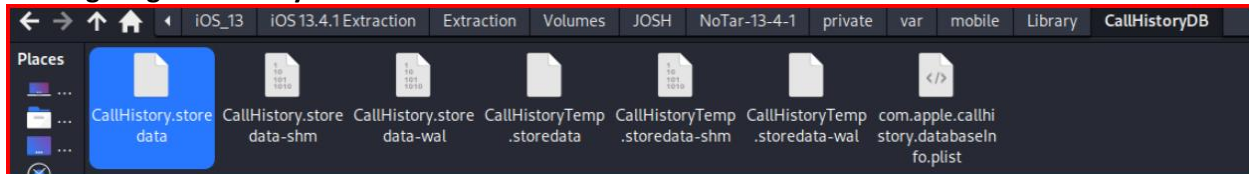
ABMultiValue

Table: ABMultiValue

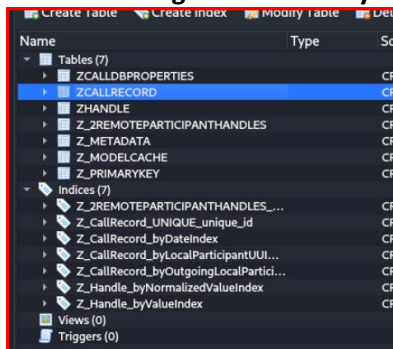
UID	record_id	property	identifier	label	value	guid
F...	Filter	Filter	Filter	Filter	Filter	Filter
1	4	1	3	0	1 (919) 579-0479	A17B9A21-27DE-4D28-ABC3-DFE251369F55
2	5	1	3	1	2 (919) 391-2507	CCAD8E14-D4A0-494E-A473-E9CAC7331D64
3	6	1	4	0	NULL joshua.hickman1@me.com	1C393A4D-A177-4E77-9441-E0C3A823424C

Investigating Call logs

Investigating Call History Information



Understanding the call history database and tables



Academic Integrity and Authorization Statement

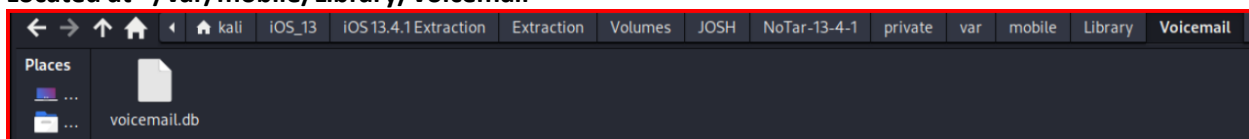
I Favour Madu confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.

Examining call history database structure

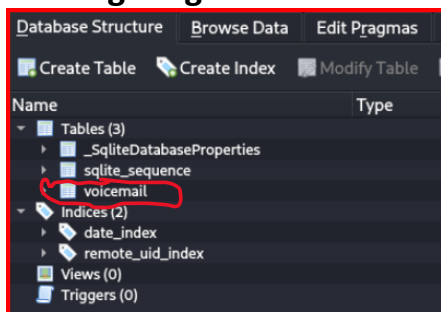
ZCALLBPROPERTIES		ZCALLRECORD									
ZCALLRECORD											
Table ZCALLRECORD											
Filter in any col...											
Z_PN	Z_INT	Z_OUT	ANSWERED	ZCALL_CATEGORY	ZCALLTYPE	ZDISCONNECTED_CAUSE	ZFACE_TIME	DATA	ZFILTER		
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter		
1	12	2	2	0	1	1	0		NALL		
2	16	2	2	0	1	1	0		NALL		
3	26	2	2	0	1	1	0		NALL		
4	27	2	2	0	1	1	0		NALL		
5	28	2	2	0	1	1	0		NALL		
6	29	2	2	0	1	1	0		NALL		
7	30	2	2	0	1	1	0		NALL		
8	31	2	1	0	1	1	0		NALL		
9	32	2	2	0	1	1	0		NALL		
10	33	2	2	0	1	1	0		NALL		
11	34	2	2	0	1	1	0		NALL		
12	35	2	2	0	1	1	0		NALL		
13	36	2	2	0	1	1	0		NALL		
14	37	2	2	0	1	1	0		NALL		
15	38	2	2	0	1	1	0		NALL		
16	39	2	2	0	1	1	0		NALL		

Investigating Voicemails

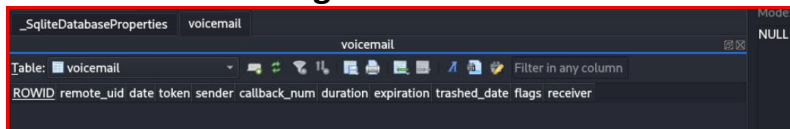
Located at ~/var/mobile/Library/Voicemail



Investigating voicemail table



No voicemail message



Calendar Investigation

Calendar scenario

Assume that suspect added an event along with the attendees for any date and added a reminder about that event

Questions

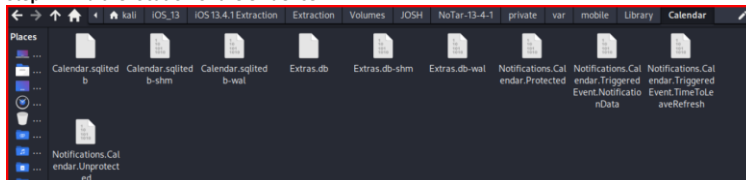
What is the date of the scheduled event?

Who are the attendees?

For which date is the reminder set?

Are there any other events saved by the suspect?

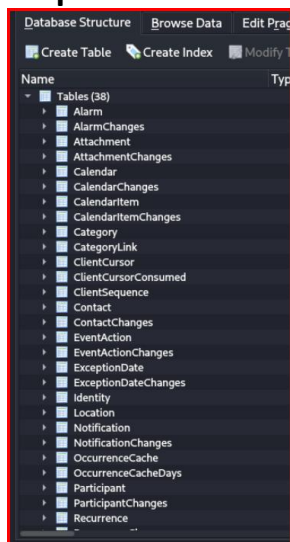
Step 1: Find the location of the evidence



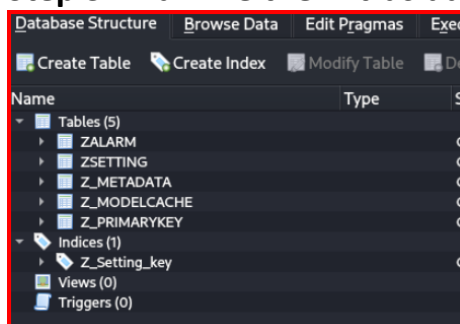
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Step 2: Examine the Calendar database



Step 3: Examine the Extras database

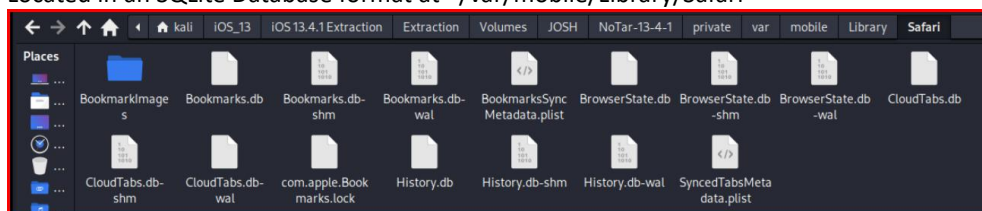


What is the date of the scheduled event? Calendar Item table
Who are the attendees? Participant table
For which date is the reminder set? Calendar Item table
Are there any other events saved by the suspect? Calendar table

Safari Investigation

How to find evidence generated by Safari?

Located in an SQLite Database format at ~/var/mobile/Library/Safari



Bookmarks.db – Safari bookmarks, i.e., saved URLs.

Browserstate.db – Stores active browser tabs in the order they were opened

Cloudtabs.db – Stores browser tabs to be carried onto another apple device.

For context - This feature automatically syncs open tabs across devices with the same iCloud account. For example, you can view a list of all tabs open on your MacBook while using Safari on the iPhone or iPad.

History.db – Stores browser history, can be cleared if user clears history.

SAFARI SENARIO STEP BY STEP SOLUTION

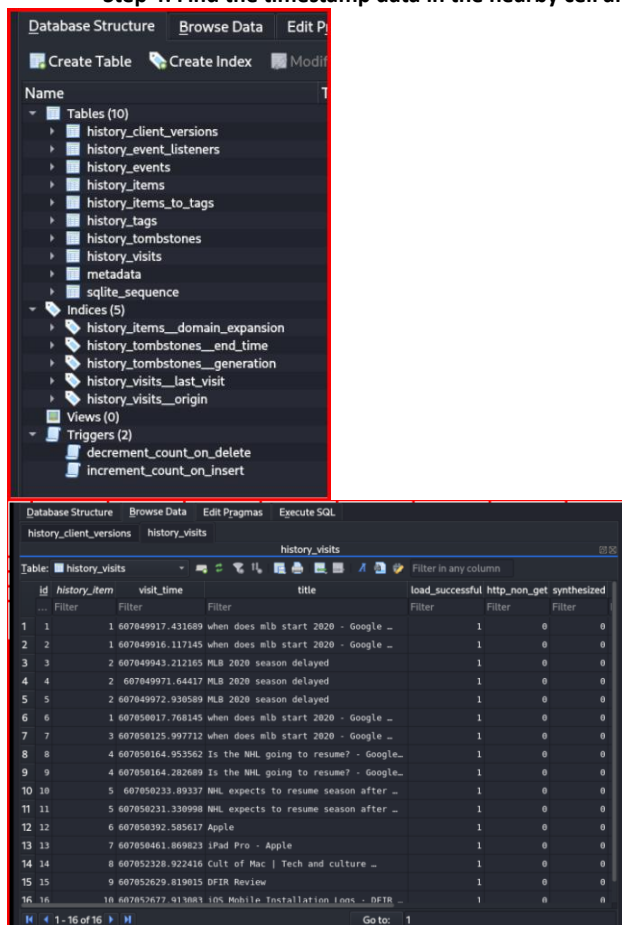
Step 1: Find the database where browser history is stored

- Step 2: Find the table contains the browser history data**
- Step 3: Find the column contains the search data**

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- **Step 4: Find the timestamp data in the nearby cell and convert**

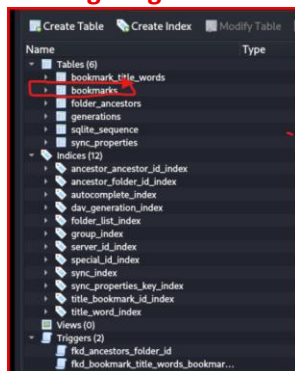


The screenshot shows a database interface with a table named 'history_visits'. The table has the following columns: id, history_item, visit_time, title, load_successful, http_non_get, synthesized. The data is as follows:

id	history_item	visit_time	title	load_successful	http_non_get	synthesized
1	1	607049917.431689	when does mlb start 2020 - Google ..	1	0	0
2	1	607049916.117145	when does mlb start 2020 - Google ..	1	0	0
3	3	2 607049943.212165	MLB 2020 season delayed	1	0	0
4	2	607049971.64417	MLB 2020 season delayed	1	0	0
5	5	2 607049972.938589	MLB 2020 season delayed	1	0	0
6	6	1 607058017.760145	when does mlb start 2020 - Google ..	1	0	0
7	7	3 607058125.997712	when does mlb start 2020 - Google ..	1	0	0
8	4	607058164.953562	Is the NHL going to resume? - Google..	1	0	0
9	4	607058164.282689	Is the NHL going to resume? - Google..	1	0	0
10	10	5 607058233.89337	NHL expects to resume season after ..	1	0	0
11	5	607058231.330998	NHL expects to resume season after ..	1	0	0
12	12	6 607058392.585617	Apple	1	0	0
13	13	7 607058461.869823	iPad Pro - Apple	1	0	0
14	14	8 607052328.922416	Cult of Mac Tech and culture ..	1	0	0
15	9	607052629.819015	DFIR Review	1	0	0
16	16	10 607052677.913083	iOS Mobile Installation Issues - DFIR ..	1	0	0

We can determine that the phone user searched two strings, When does MLB start 2020 & Is the NHL going to resume? Between 12:58am and 1:02am 3/28/2020 GMT.

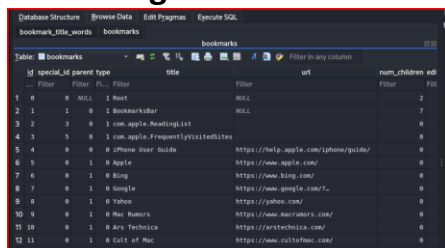
Investigating bookmarks.db



The screenshot shows a database interface with a table named 'bookmarks'. The table has the following columns: id, special_id, parent_type, title, url, num_children. The data is as follows:

id	special_id	parent_type	title	url	num_children
1	0	0	Root	NULL	2
2	1	0	BookmarksBar	NULL	7
3	2	0	1 new apple-headling list	NULL	0
4	3	0	1 new apple frequent visited sites	NULL	0
5	4	0	1 iPhone user guide	https://help.apple.com/iphone/guide/	0
6	5	0	1 Apple	https://www.apple.com/	0
7	6	0	1 Bing	https://www.bing.com/	0
8	7	0	1 Google	https://www.google.com/	0
9	8	0	1 Yahoo	https://yaho.com/	0
10	9	0	1 Mac Rumors	https://www.macrumors.com/	0
11	10	0	1 Ars Technica	https://arstechnica.com/	0
12	11	0	1 Cult of Mac	https://www.cultofmac.com/	0

Examining Bookmark Table



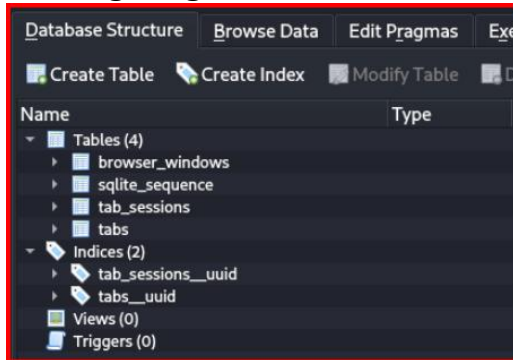
The screenshot shows a database interface with a table named 'bookmarks'. The table has the following columns: id, special_id, parent_type, title, url, num_children. The data is as follows:

id	special_id	parent_type	title	url	num_children
1	0	0	Root	NULL	2
2	1	0	BookmarksBar	NULL	7
3	2	0	1 new apple-headling list	NULL	0
4	3	0	1 new apple frequent visited sites	NULL	0
5	4	0	1 iPhone user guide	https://help.apple.com/iphone/guide/	0
6	5	0	1 Apple	https://www.apple.com/	0
7	6	0	1 Bing	https://www.bing.com/	0
8	7	0	1 Google	https://www.google.com/	0
9	8	0	1 Yahoo	https://yaho.com/	0
10	9	0	1 Mac Rumors	https://www.macrumors.com/	0
11	10	0	1 Ars Technica	https://arstechnica.com/	0
12	11	0	1 Cult of Mac	https://www.cultofmac.com/	0

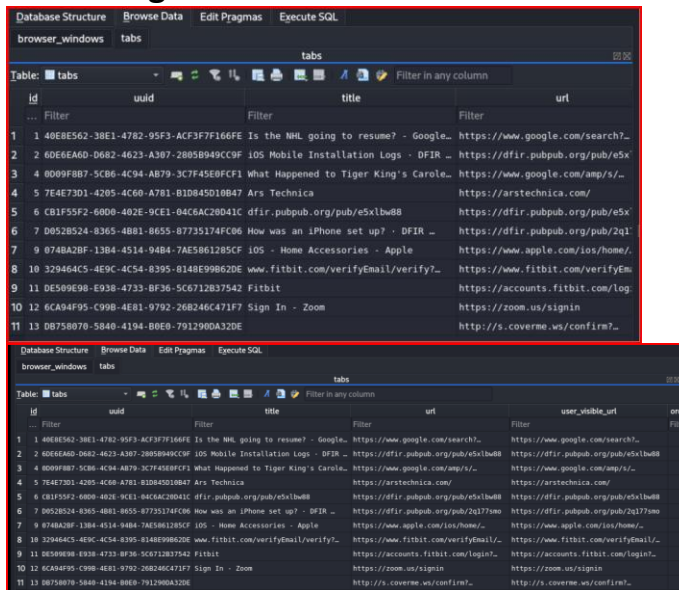
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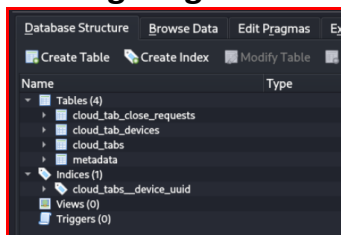
Investigating BrowserState.db



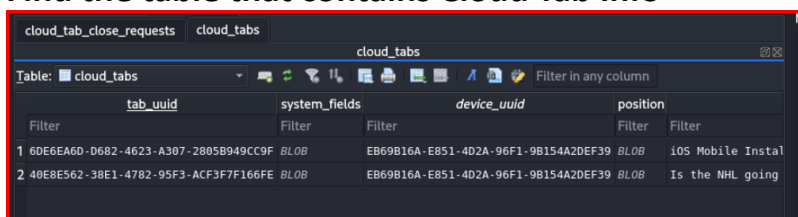
Examining Browser Tabs Table



Investigating Cloud Tabs



Find the table that contains Cloud Tab Info



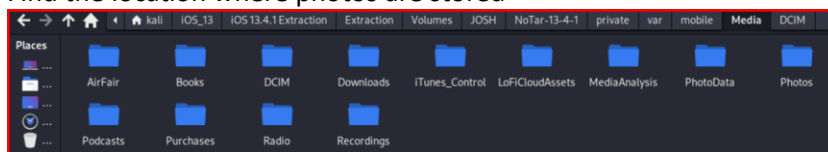
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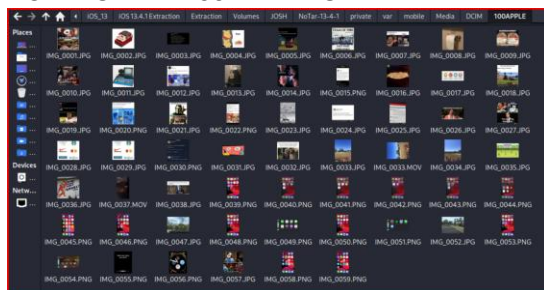
Photos Investigation

- The Photos app is where all the photos you take and save with your iOS devices live.

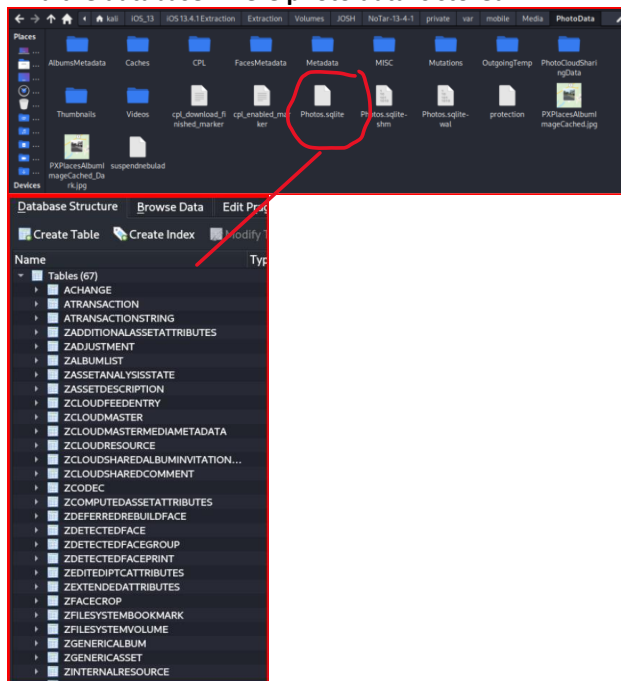
Find the location where photos are stored



DCIM FOLDER-100APPLE FOLDER



Find the database where photo data is stored



Investigating ZADDITIONALASSETATTRIBUTES

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Database Structure Browse Data Edit Pragas Execute SQL									
ACHANGE ZADDITIONALASSETATTRIBUTES ZADDITIONALASSETATTRIBUTES									
Table: ZADDITIONALASSETATTRIBUTES									
Z_PK	Z_ENT	Z_OPT	Z_FOPT	Z_P1	Z_P2	Z_P3	Z_P4	Z_P5	Z_P6
43	42	1	14	1	0	0	0	0	0
44	44	1	14	1	0	0	0	0	0
45	45	1	13	1	0	0	0	0	0
46	46	1	16	1	0	0	0	0	0
47	47	1	19	0	0	0	0	0	0
48	48	1	13	0	0	0	0	0	0
49	49	1	22	1	0	0	0	0	0
50	50	1	13	0	0	0	0	0	0
51	51	1	20	1	0	0	0	0	0
52	52	1	18	1	0	0	0	0	0
53	53	1	14	0	0	0	0	0	0
54	54	1	23	1	0	0	0	0	0
55	55	1	13	0	0	0	0	0	0
56	56	1	14	1	0	0	0	0	0
57	57	1	14	1	0	0	0	0	0
58	58	1	11	1	0	0	0	0	0

Database Structure Browse Data Edit Pragas Execute SQL									
ACHANGE ZADDITIONALASSETATTRIBUTES ZADDITIONALASSETATTRIBUTES									
Table: ZADDITIONALASSETATTRIBUTES									
DAI	ZSCENANALYSIS	ZTIMESTAMP	ZFIR	ZCREATOR	ZBUNDLEID	ZEXIFTIMESTAMP	ZSTRING	ZFSE	ZMASTERFINGERPRINT
23	N.	606941715.140196	N.	com.reddit.Reddit	N.	N.	N.	N.	IM6023.JPG
24	N.	606941884.166597	N.	com.reddit.Reddit	N.	N.	N.	N.	IM6024.JPG
25	N.	606942179.7701	N.	com.reddit.Reddit	N.	N.	N.	N.	IM6025.JPG
26	N.	606942246.169049	N.	com.reddit.Reddit	N.	N.	N.	N.	IM6026.JPG
27	N.	606942615.163699	N.	com.reddit.Reddit	N.	N.	N.	N.	IM6027.JPG
28	N.	606942280	N.	N.	N.	N.	N.	N.	c1755af3-Saf8-446d-9d97-31b
29	N.	606942280	N.	N.	N.	N.	N.	N.	c1755af3-Saf8-446d-9d97-31b
30	N.	606944802.435647	N.	com.silentcircle.SilentP	N.	N.	N.	N.	IM6030.JPG
31	N.	607806000.6531	N.	de.tutan.tutanota	N.	N.	N.	N.	IM6031.JPG
32	N.	607807637.815699	N.	ch.protonmail.protonmail	N.	N.	N.	N.	IM6032.JPG
33	N.	608328569.281667	N.	N.	N.	2020:04:11 13:56:09	N.	N.	IM6033.JPG
34	N.	608328615.866572	N.	N.	N.	2020:04:11 13:56:56	N.	N.	IM6034.JPG
35	N.	608345100.196669	N.	us.zoom.videomeetings	N.	N.	N.	N.	IM6035.JPG
36	N.	608347752.244466	N.	com.mysckr.wickr	N.	N.	N.	N.	IM6036.JPG
37	N.	608349429	N.	N.	N.	2020:04:11 21:57:09	N.	N.	f676a0749a13b99a7209d4cc053
38	N.	608349185.156771	N.	com.annex.MobileSMS	N.	N.	N.	N.	IM6038.JPG

Database Structure Browse Data Edit Pragas Execute SQL									
ACHANGE ZADDITIONALASSETATTRIBUTES ZADDITIONALASSETATTRIBUTES									
Table: ZADDITIONALASSETATTRIBUTES									
ZPUBLICGLOBALUID	ZORIGINFILE	ZORIGINFILESIZE	ZORIGINFILEHEIGHT	ZORIGINFILEWIDTH	ZORIGINFILEMIME	ZORIGINFILEEXT	ZORIGINFILETYPE	ZORIGINFILESUBTYPE	ZORIGINFILECATEGORY
38	N.	N.	N.	N.	N.	N.	N.	N.	N.
39	N.	N.	N.	N.	N.	N.	N.	N.	N.
40	N.	N.	N.	N.	N.	N.	N.	N.	N.
41	N.	N.	N.	N.	N.	N.	N.	N.	N.
42	N.	N.	N.	N.	N.	N.	N.	N.	N.
43	N.	N.	N.	N.	N.	N.	N.	N.	N.
44	N.	N.	N.	N.	N.	N.	N.	N.	N.
45	N.	N.	N.	N.	N.	N.	N.	N.	N.
46	N.	N.	N.	N.	N.	N.	N.	N.	N.
47	N.	N.	N.	N.	N.	N.	N.	N.	N.
48	N.	N.	N.	N.	N.	N.	N.	N.	N.
49	N.	N.	N.	N.	N.	N.	N.	N.	N.
50	N.	N.	N.	N.	N.	N.	N.	N.	N.
51	N.	N.	N.	N.	N.	N.	N.	N.	N.
52	N.	N.	N.	N.	N.	N.	N.	N.	N.
53	N.	N.	N.	N.	N.	N.	N.	N.	N.
54	N.	N.	N.	N.	N.	N.	N.	N.	N.
55	N.	N.	N.	N.	N.	N.	N.	N.	N.
56	N.	N.	N.	N.	N.	N.	N.	N.	N.
57	N.	N.	N.	N.	N.	N.	N.	N.	N.
58	N.	N.	N.	N.	N.	N.	N.	N.	N.

Investigating ZGENERICASSET

ACHANGE ZGENERICASSET									
Table: ZGENERICASSET									
T	I	C	H	E	L	I	N	C	O
1	8	0	0	0	N.	N.	N.	N.	1
2	3	0	0	0	N.	N.	N.	N.	1
3	0	0	0	0	N.	N.	N.	N.	1
4	7	0	0	0	N.	N.	N.	N.	1
5	7	0	0	0	N.	N.	N.	N.	1
6	7	0	0	0	N.	N.	N.	N.	1
7	3	0	0	0	N.	N.	N.	N.	1
8	9	0	0	0	N.	N.	N.	N.	1
9	6	0	0	0	N.	N.	N.	N.	1
10	0	0	0	0	N.	N.	N.	N.	1
11	5	0	0	0	N.	N.	N.	N.	1
12	6	0	0	0	N.	N.	N.	N.	1
13	6	0	0	0	N.	N.	N.	N.	1
14	9	0	0	0	N.	N.	N.	N.	1
15	9	0	0	0	N.	N.	N.	N.	1
16	4	0	0	0	N.	N.	N.	N.	1

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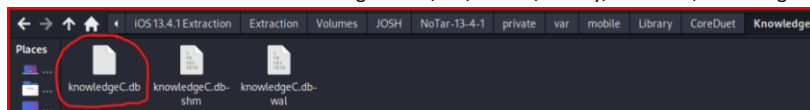
ZGENERICSSET													
IT	ZADDEDDATE	ZJUSTMENTTIMESTAMP	ZANALYSISSTATE	ZMODIFICATIONDATE	HPWEC	ERI	ZCURATIONSORE	ZDATECREATED	URATH	ZFACEADJUSTMENTV			
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter			
1	606575124.564286	NUL	606598187.841724	N.	NUL	N.	0.25	606575124.541041	0.0	606575124.541041			
2	606575139.765026	NUL	607878918.397668	N.	NUL	N.	0.75	606575139.762112	0.0	606575139.762112			
3	606575450.328813	NUL	608425651.185999	N.	NUL	N.	0.5	606575450.327284	0.0	606575450.327284			
4	606576321.20214	NUL	606598187.84289	N.	NUL	N.	0.25	606576302.969417	0.0	606576302.969417			
5	606576321.291399	NUL	607878918.400772	N.	NUL	N.	0.5	606576302.678356	0.0	606576302.678356			
6	606576321.424513	NUL	607878918.399666	N.	NUL	N.	0.5	606576302.734282	0.0	606576302.734282			
7	606576321.565905	NUL	607878918.401176	N.	NUL	N.	0.5	606576302.783361	0.0	606576302.783361			
8	606576321.57976	NUL	606598187.845094	N.	NUL	N.	0.25	606576302.81583	0.0	606576302.81583			
9	60657621.964396	NUL	606598187.845683	N.	NUL	N.	0.25	606576663.109	0.0	606576663.109			
10	606577622.074259	NUL	606598187.84611	N.	NUL	N.	0.25	606577214.935	0.0	606577214.935			
11	606578589.270193	NUL	607878918.397208	N.	NUL	N.	0.5	606578589.257008	0.0	606578589.257008			
12	606582401.930913	NUL	606598187.84714	N.	NUL	N.	0.5	606582401.125989	0.0	606582401.125989			
13	606582495.178322	NUL	607878918.398467	N.	NUL	N.	0.5	606582494.159426	0.0	606582494.159426			
14	606582557.683429	NUL	606598187.848171	N.	NUL	N.	0.25	606582556.250668	0.0	606582556.250668			
15	606595332.007215	606597676.665731	606598136.025578	N.	NUL	N.	0.25	606595331	0.0	606597676.665731			
16	606595431.201642	NUL	606598136.026072	N.	NUL	N.	0.25	606595431.104936	0.0	606595431.104936			

ZGENERICSSET													
HA	ZDIRECTORY	ZFILENAME	POIR	ZORIGINALCOLORSPACE	ZUNIFORMIDENTIFIER	ZUID	ZIMAG						
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter						
1	N.	DCIM\100APPLE	IMG_0001.JPG	N.	N.	Display P3	public.jpeg	6F8C71F7-92AE-42D0-8FC2-1B489FC879E3	0.0/0				
2	N.	DCIM\100APPLE	IMG_0002.JPG	N.	N.	Adobe RGB (1998)	public.jpeg	B1C6C0C3-E697-4F2A-8C23-588942C39EC	0.0/0				
3	N.	DCIM\100APPLE	IMG_0003.JPG	N.	N.	Display P3	public.jpeg	265EB474-2490-4A7C-ABDC-438BA1C38A2	0.0/0				
4	N.	DCIM\100APPLE	IMG_0004.JPG	N.	N.	Display P3	public.jpeg	E4ED88F5-733F-4929-9757-0669C8F06782	0.0/0				
5	N.	DCIM\100APPLE	IMG_0005.JPG	N.	N.	Display P3	public.jpeg	9518AA2-6957-4F36-9616-3F84D9829608	0.0/0				
6	N.	DCIM\100APPLE	IMG_0006.JPG	N.	N.	Display P3	public.jpeg	62E810A5-1A6A-4550-83D3-268DF5531AA4	0.0/0				
7	N.	DCIM\100APPLE	IMG_0007.JPG	N.	N.	Display P3	public.jpeg	E5E42C05-1E18-4586-98EF-777C88AAC15	0.0/0				
8	N.	DCIM\100APPLE	IMG_0008.JPG	N.	N.	Display P3	public.jpeg	F4E94CE7-8216-4B3A-99F1-90F285E52A81	0.0/0				
9	N.	DCIM\100APPLE	IMG_0009.JPG	N.	N.	sRGB IEC61966-2.1	public.jpeg	9086954C-A754-4711-982F-8AC805648026	0.0/0				
10	N.	DCIM\100APPLE	IMG_0010.JPG	N.	N.	sRGB IEC61966-2.1	public.jpeg	E277C806-2914-403A-AC91-86808371E3CA	0.0/0				
11	N.	DCIM\100APPLE	IMG_0011.JPG	N.	N.	sRGB IEC61966-2.1	public.jpeg	0C43D2A5-B051-4E19-AB14-2A0505A23308	0.0/0				
12	N.	DCIM\100APPLE	IMG_0012.JPG	N.	N.	Display P3	public.jpeg	0A720A08-8E5C-4748-9F71-3A772978E085	0.0/0				
13	N.	DCIM\100APPLE	IMG_0013.JPG	N.	N.	Display P3	public.jpeg	F2A831C4-0A27-474C-887C-803E40F4218C	0.0/0				
14	N.	DCIM\100APPLE	IMG_0014.JPG	N.	N.	sRGB IEC61966-2.1	public.jpeg	6385982F-8889-4B1A-8A45-AB9A4C827286	0.0/0				
15	N.	DCIM\100APPLE	IMG_0015.PNG	N.	N.	sRGB IEC61966-2.1	public.jpeg	7887385C-871A-470F-91FA-0A544E287872	0.0/0				
16	N.	DCIM\100APPLE	IFR_0016.JPG	N.	N.	sRGB IEC61966-2.1	public.jpeg	8FF8F75A-800F-497A-9019-70F48000F87E	0.0/0				

KnowledgeC Investigation

Querying within KnowledgeC.db

Database Location- Navigate to `~/var/mobile/Library/CoreDuet/Knowledge`



Database structure

Tables (16)
▸ ZADDITIONCHANGESSET
▸ ZCONTEXTUALCHANGEREISTRATION
▸ ZCONTEXTUALKEYPATH
▸ ZCUSTOMMETADATA
▸ ZDELETIONCHANGESSET
▸ ZHISTOGRAM
▸ ZHISTOGRAMVALUE
▸ ZKEYVALUE
▸ ZOBJECT
▸ ZSOURCE
▸ ZSTRUCTUREDMETADATA
▸ ZSYNCPER
▸ Z_4EVENT
▸ Z_METADATA
▸ Z_MODELCACHE
▸ Z_PRIMARYKEY
Indices (23)
▸ ZCUSTOMMETADATA_ZOBJECT_IND...
▸ ZHISTOGRAMVALUE_ZHISTOGRAM_...
▸ ZOBJECT_ZEVENT_INDEX
▸ ZOBJECT_ZOBJECT_INDEX
▸ ZOBJECT_ZSOURCE_INDEX
▸ ZOBJECT_ZSTRUCTUREDMETADATA...
▸ ZOBJECT_ZSUBJECT_INDEX
▸ ZOBJECT_ZVALUE_INDEX
▸ ZOBJECT_Z_ENT_INDEX
▸ Z_4EVENT_Z_11EVENT_INDEX
▸ Z_CustomMetadata_byNameIndex

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I Favour Madu confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.

Querying with SQL

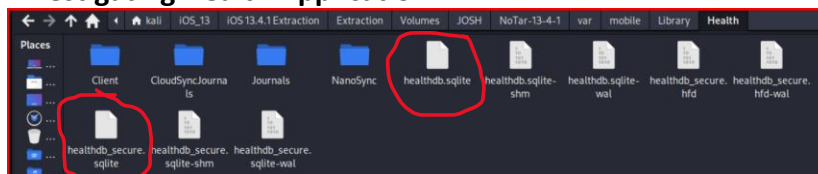
The screenshot displays a database interface with a table named ZOBJECT. The table has columns: Z_PK, Z_ENT, Z_OPT, ZUIDHASH, ZEVENT, and ZSOURCE. The data is filtered to show 16 rows. Below the table, an SQL query is entered in the 'Execute SQL' tab. The query is as follows:

```

1 SELECT
2   datetime(ZOBJECT.ZCREATIONDATE+978307200, 'UNIXEPOCH', 'LOCALTIME') as "ENTRY_CREATION",
3   CASE ZOBJECT.ZSTARTDAYOFWEEK
4     WHEN "1" THEN "Sunday"
5     WHEN "2" THEN "Monday"
6     WHEN "3" THEN "Tuesday"
7     WHEN "4" THEN "Wednesday"
8     WHEN "5" THEN "Thursday"
9     WHEN "6" THEN "Friday"
10    WHEN "7" THEN "Saturday"
11  END "DAY OF WEEK",
12   ZOBJECT.ZSECONDSFROMGMT/3600 AS "GMT OFFSET",
13   datetime(ZOBJECT.ZSTARTDATE+978307200, 'UNIXEPOCH', 'LOCALTIME') as "START",
14   datetime(ZOBJECT.ZENDDATE+978307200, 'UNIXEPOCH', 'LOCALTIME') as "END",
15   (ZOBJECT.ZENDDATE - ZOBJECT.ZSTARTDATE) as "USAGE IN SECONDS",
16   ZOBJECT.ZSTREAMNAME,
17   ZOBJECT.ZVALUESTRING
18 FROM ZOBJECT

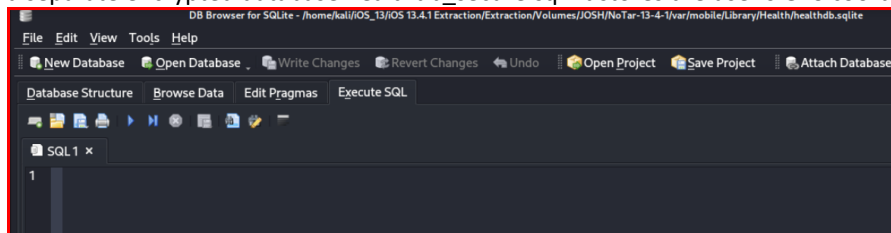
```

Investigating Health Application



HealthDB and HealthDB_Secure?

The two databases contain all information collected or received by the Health app with one exception. There is also a separate encrypted database healthdb_secure.sql. It stores the user's GPS coordinates collected during workouts



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International Cybersecurity and Digital Forensics Academy (ICDFA)

```
SQL 1* x
1 select datetime (creation_date+978307200, 'unixepoch','localtime') as timestamp,
2 name, manufacturer, model, hardware, software from source_devices
3
4
```

	timestamp	name	manufacturer	model	hardware	software
1	2020-03-21 18:33:20	iPhone	Apple Inc.	iPhone	iPhone8,4	13.3.1
2	2020-03-21 18:33:20	__NONE__				
3	2020-03-21 16:28:48	Apple Watch	Apple Inc.	Watch	Watch4,3	6.1.3
4	2020-03-21 16:28:48	__NONE__				
5	2020-03-27 15:06:14	AirPods	Apple Inc.	0x2002		
6	2020-03-31 15:08:08	AirPods Pro	Apple Inc.	0x200e		

Execution finished without errors.
Result: 8 rows returned in 12ms
At line 1:
select datetime (creation_date+978307200, 'unixepoch','localtime') as timestamp,
name, manufacturer, model, hardware, software from source_devices

Health Sources – Applications - Statement

```
SQL 1* x  SQL 2* x
1 select name, bundle_id, source_options, local_device, product_type from sources
2
```

	name	bundle_id	source_options	local_device	product_type
1	This Is's iPhone	com.apple.health.284D8298-B83A-422F-...	2	1	iPhone8,4
2	Health	com.apple.Health	3	0	
3	This Is's Apple Watch	com.apple.health.8D0D238A-F8E1-4102-...	2	0	Watch4,3
4	This Is's Apple Watch	com.apple.health.9676B92F-9138-4B9D-...	2	0	Watch4,3
5	This Is's Apple Watch	com.apple.health.9676B92F-9138-4B9D-...	2	0	Watch4,3
6	Sync Solver	com.redshiftdev.Fitbit-Health-Sync	5	0	

Execution finished without errors.
Result: 6 rows returned in 10ms
At line 1:
select name, bundle_id, source_options, local_device, product_type from sources

```
SQL 1* x
1 SELECT datetime(energy_burned_goal_date+978307200, 'unixepoch','utc') as "Energy Burned Goal Date",
2 datetime(cache_index+978307200, 'unixepoch','utc') as "cache_index", energy_burned, energy_burned_goal,
3 active_hours, active_hours_goal, brisk_minutes, brisk_minutes_goal, steps, walk_distance
4 from activity_caches
5
```

	Energy Burned Goal Date	cache_index	energy_burned	energy_burned_goal	active_hours	active_hours_goal	brisk_minutes	brisk_minutes_goal	steps	walk_dis
1	0000-12-30 05:00:00	2020-03-20 04:00:00	0.0	0.0	0.0	12.0	0.0	30.0	0.0	0.0
2	2020-03-22 04:51:53	2020-03-21 04:00:00	28.965	480.0	1.0	12.0	0.0	30.0	54.0	45.659500H
3	2020-03-22 04:51:53	2020-03-22 04:00:00	454.2110000000001	480.0	15.0	12.0	28.0	30.0	5288.0	4141.5635
4	2020-03-22 04:51:53	2020-03-23 04:00:00	597.5709999999999	480.0	14.0	12.0	53.0	30.0	7930.0	6242.1125
5	2020-03-22 04:51:53	2020-03-24 04:00:00	525.601	480.0	14.0	12.0	40.0	30.0	5951.0	4704.2290
6	2020-03-22 04:51:53	2020-03-25 04:00:00	472.9620000000001	480.0	13.0	12.0	28.0	30.0	5560.0	4387.4990

Execution finished without errors.
Result: 21 rows returned in 10ms
At line 1:
SELECT datetime(energy_burned_goal_date+978307200, 'unixepoch','utc') as "Energy Burned Goal Date",
datetime(cache_index+978307200, 'unixepoch','utc') as "cache_index", energy_burned, energy_burned_goal,
active_hours, active_hours_goal, brisk_minutes, brisk_minutes_goal, steps, walk_distance
from activity_caches

ACTIVITY CACHE RESULT

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International Cybersecurity and Digital Forensics Academy (ICDFA)

SQL 1* x

```
1 SELECT datetime(energy_burned_goal_date+978307200,'unixepoch','utc') as "Energy Burned Goal Date",
2 datetime(cache_index+978307200,'unixepoch','utc') as "cache_index", energy_burned, energy_burned_goal,
3 active_hours, active_hours_goal, brisk_minutes, brisk_minutes_goal, steps, walk_distance
4 from activity_caches
5
```

	Energy Burned Goal Date	cache_index	energy_burned	energy_burned_goal	active_hours	active_hours_goal	brisk_minutes	brisk_minutes_goal	steps	walk_dis
8	2020-03-22 04:51:53	2020-03-27 04:00:00	757.9350000000001	480.0	14.0	12.0	63.0	30.0	10094.0	8785.41520
9	2020-03-22 04:51:53	2020-03-28 04:00:00	673.801	480.0	14.0	12.0	44.0	30.0	10367.0	9052.75620
10	2020-03-22 04:51:53	2020-03-29 04:00:00	534.149	480.0	15.0	12.0	37.0	30.0	7762.0	6134.26840
11	2020-03-22 04:51:53	2020-03-30 04:00:00	881.534	480.0	16.0	12.0	77.0	30.0	12580.0	11142.2070
12	2020-03-22 04:51:53	2020-03-31 04:00:00	841.444	480.0	16.0	12.0	67.0	30.0	11119.0	9976.22440
13	2020-03-22 04:51:53	2020-04-01 04:00:00	576.732	480.0	15.6	12.0	44.0	30.0	7749.0	6229.00690
14	2020-03-22 04:51:53	2020-04-03 04:00:00	753.338	480.0	11.0	12.0	62.0	30.0	8787.0	8299.91420
15	2020-03-22 04:51:53	2020-04-02 04:00:00	812.379	480.0	14.0	12.0	65.0	30.0	10953.0	10099.5110
16	2020-03-22 04:51:53	2020-04-11 04:00:00	0.0	480.0	0.0	12.0	0.0	30.0	0.0	0.0
17	2020-03-22 04:51:53	2020-04-12 04:00:00	196.67	480.0	6.0	12.0	13.0	30.0	1633.0	1323.20710
18	2020-03-22 04:51:53	2020-04-13 04:00:00	599.029	480.0	14.0	12.0	35.0	30.0	6642.0	5423.58330
19	2020-03-22 04:51:53	2020-04-14 04:00:00	631.4399999999999	480.0	16.0	12.0	34.0	30.0	5970.0	4766.43760
20	2020-03-22 04:51:53	2020-04-15 04:00:00	625.1560000000001	480.0	14.0	12.0	37.0	30.0	7442.0	6007.91280
21	2020-03-22 04:51:53	2020-04-16 04:00:00	253.141	480.0	7.0	12.0	8.0	30.0	2657.0	2990.75640

Workouts – Statement

```
select samples.data_id, datetime(samples.start_date+978307200,'unixepoch','localtime') as "Start Date",
datetime(samples.end_date+978307200,'unixepoch','localtime') as "End Date", data_type, activity_type,
total_energy_burned, total_basal_energy_burned, total_distance, goal_type, goal FROM workouts
join samples on workouts.data_id = samples.data_id
```

SQL 1* x SQL 2* x

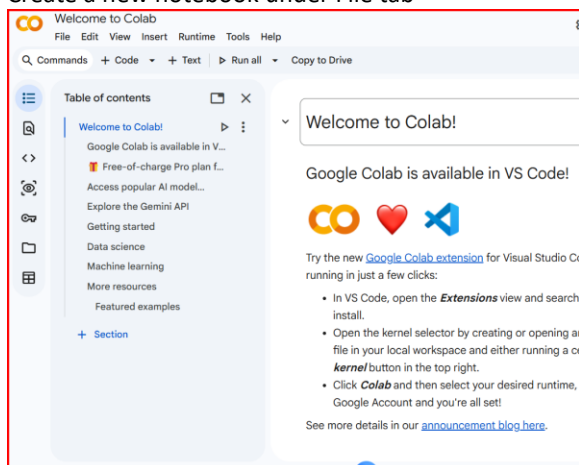
```
2 datetime(samples.end_date+978307200,'unixepoch','localtime') as "End Date", data_type, activity_type,
3 total_energy_burned, total_basal_energy_burned, total_distance, goal_type, goal FROM workouts
4 join samples on workouts.data_id = samples.data_id
5
```

	data_id	Start Date	End Date	data_type	activity_type	total_energy_burned	total_basal_energy_burned	total_distance	goal_type	goal
1	15017	2020-03-27 14:57:39	2020-03-27 15:15:39	79	37	208.57206077784	26.3355178817357	3.24798000599684	0	NULL
2	18146	2020-03-28 10:41:25	2020-03-28 10:58:44	79	37	203.107029922126	25.3377903447887	3.23536024568131	0	NULL
3	24649	2020-03-30 08:44:04	2020-03-30 09:10:35	79	37	302.580231726703	38.851842234397	4.84550446943786	0	NULL
4	30219	2020-03-31 14:58:47	2020-03-31 15:24:59	79	37	302.861406554144	38.3217232399215	4.86484843934411	0	NULL
5	36805	2020-04-02 13:36:41	2020-04-02 14:02:31	79	37	301.889197537766	37.9672273635361	4.85304426234419	0	NULL
6	44801	2020-04-03 15:11:28	2020-04-03 15:37:49	79	37	369.850771357538	46.0088782026736	4.86274976773959	0	NULL

Creating Simple Graphs with Python

Visit Google Colab- <https://colab.research.google.com/>

Create a new notebook under File tab

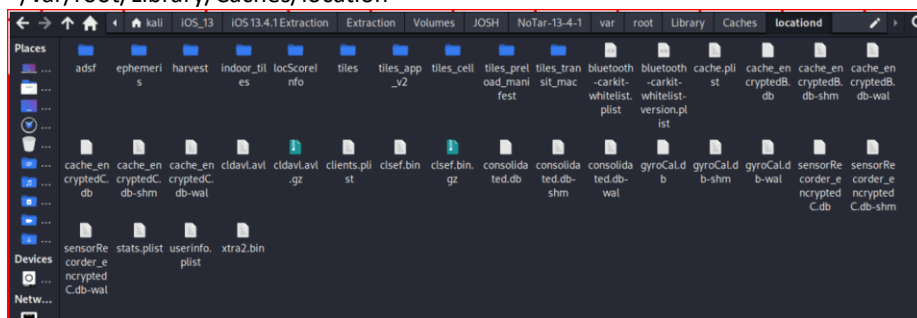


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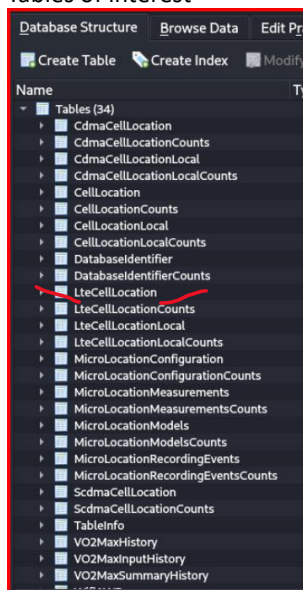
I **Favour Madu** confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.

iOS Location Forensics

~/var/root/Library/Caches/location



Tables of interest



Conclusion

This examination demonstrated a structured approach to analysing an iPhone forensic image running iOS 13.4.1. Using Kali Linux within VMware Workstation, the extraction was obtained, its integrity was checked with MD5 and SHA-256 hashes, and the file system was examined for relevant property lists, SQLite databases, application containers, and system artefacts. The process showed how command-line utilities, property-list parsing, database review, and Apple timestamp conversion can be combined to identify and interpret mobile evidence.

The confirmed findings include the registered telephone number and key subscriber and network identifiers, including the ICCID, IMSI, IMEI, MCC, and MNC. The examination also identified the principal storage locations for Messages, contacts, call history, voicemail, calendars, Safari, photos, KnowledgeC, Health, and location artefacts. Safari history showed searches concerning the start of the 2020 MLB season and the possible resumption of the NHL between approximately 00:58 and 01:02 GMT on 28 March 2020. Examination of the voicemail data did not identify a voicemail message.

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These results confirm that native iOS artefacts can support reconstruction of device configuration and selected user activity when their source paths, database structures, and timestamps are interpreted carefully. However, findings should be treated as artefact-based observations rather than proof of who personally performed an action. A defensible final opinion therefore requires correlation across independent sources, validation with another forensic method or tool, and complete documentation of evidence handling and analytical decisions.

Limitations

The conclusions in this report are subject to the following limitations:

- **Scope of acquisition:** The examination was based on a published extraction rather than a live-device acquisition. Artefacts absent from the extraction, encrypted content, cloud-only data, and data removed before acquisition may therefore be unavailable.
- **Incomplete artefact analysis:** Several sections identify databases, tables, or investigative steps without recording final results. These include aspects of calendars, contacts, call history, photos, KnowledgeC, Health, location services, account information, device serial and model details, and jailbreak indicators.
- **Tool and parsing limitations:** Some databases produced errors in SQLite Browser, and no full cross-tool validation is documented. Parser behaviour, unsupported schemas, or database corruption may affect displayed values.
- **Timestamp interpretation:** Apple and Unix timestamps require conversion, and results may differ according to epoch, time zone, daylight-saving treatment, and whether a value is stored in seconds or nanoseconds. The Safari finding is reported in GMT, but other values should be normalised before correlation.
- **Attribution:** A device artefact can show that an event occurred on, or was recorded by, the device; it does not by itself establish the identity or intent of the person responsible.
- **Deleted and volatile data:** This report does not document a systematic recovery of deleted records, unallocated space, volatile memory, or transient network activity.
- **Integrity and reproducibility:** Hash values are recorded for the downloaded archive, but the report does not consistently provide hashes for every extracted database or exported result. A complete examination log and working-copy verification are required for independent reproduction.
- **Privacy and disclosure:** The report contains personal and subscriber identifiers. Distribution should be restricted, and non-essential identifiers should be redacted from teaching, assessment, or public versions.

Appendix

Appendix A: Evidence Source and Integrity

- **Evidence description:** iPhone file-system extraction running iOS 13.4.1.
- **Published archive:** ios_13_4_1.zip.

- **Extraction content:** 13.4.1.tar, iTunes Backup, and Sysdiagnose Logs.
- **Acquisition environment:** Kali Linux in VMware Workstation.
- **Recorded MD5:** 4e094801ba1e3b938b2c0cc18d0e0d1d.
- **Recorded SHA-256:**
761854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c15551.
- **Integrity note:** Recalculate and compare hashes before analysis, after transfer, and for each working copy or exported evidential file.

Appendix B: Key Artefact Locations

Artefact	Location recorded in the report	Potential evidential value
Messages	~/private/var/mobile/Library/SMS/sms.db	Messages, chats, handles, and attachments.
Safari	~/var/mobile/Library/Safari	History, bookmarks, active tabs, and cloud tabs.
Voicemail	~/var/mobile/Library/Voicemail	Voicemail records and associated media, where present.
Photos	DCIM/100APPLE and the Photos database	Media files, metadata, and asset relationships.
KnowledgeC	~/var/mobile/Library/CoreDuet/Knowledge	Device and application activity recorded in KnowledgeC.db.
Health	Health databases, including healthdb_secure.sql	Health samples, workouts, sources, and protected workout-location data.
Location	~/var/root/Library/Caches/location	Location-related records requiring contextual validation.
Application code	~/var/containers/Bundle/Application	Application bundles, signatures, frameworks, and identifiers.
Application user data	~/var/mobile/Containers/Data/Application	User-generated application data and preferences.

Appendix C: Principal Examination Commands

- **Download:** wget -q [published archive URL]
- **Confirm file details:** ls -l ios_13_4_1.zip

- **Verify archive hash:** `hashdeep ios_13_4_1.zip`
- **Extract the ZIP archive:** `unzip -q ios_13_4_1.zip`
- **Extract the TAR archive:** `tar -xvf 13-4-1.tar`
- **Parse a property list:** `plistutil -i [filename].plist`
- **Convert an Apple timestamp to Unix time:** add 978307200 seconds, then interpret the result with an explicitly recorded time zone.
- **Example Unix conversion:** `date -d @1585334532`

Appendix D: Timestamp and Reporting Notes

- Record the raw timestamp, source field, source database, conversion formula, converted value, and applied time zone.
- Confirm whether the source uses the Apple epoch beginning 1 January 2001, the Unix epoch beginning 1 January 1970, or another format.
- Check whether values are stored in seconds, milliseconds, microseconds, or nanoseconds before conversion.
- Use a consistent time standard for correlation and state clearly when a displayed value has been converted to local time.
- For every reported finding, retain the source path, relevant table and column names, query or command used, exported result, and supporting screenshot.
- Separate observed data from interpretation and record any uncertainty, conflicting artefacts, or alternative explanation.